Prototypical Spectral Basis Learning for Effective Cross-Scene Few-Shot Hyperspectral Image Classification

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***Abstract*—Existing few-shot hyperspectral image (HSI) classification methods predominantly concentrate on either mining the sparse semantic knowledge only in the target HSI with limited pixel-wise annotations or further transferring the semantic information from a source domain with the same categories. However, the semantic knowledge that can be transferred across scenes (i.e., images with diverse categories) is scarcely taken into account for performance enhancement. To mitigate this problem, we first formulate the cross-scene few-shot HSI classification (CSFS-HSIC) task and then propose a novel prototypical spectral basis learning-based CSFS-HSIC method. Specifically, we develop a prototypical spectral basis learning network (PSBLN) to extract spectral bases from an auxiliary HSI dataset with extensive pixel-wise annotations. These bases are not only interdiscriminative but also capable of accurately linearly reconstructing the prototypical spectra of all categories, enabling the network to distill cross-scene transferable semantic knowledge in the spectral domain. Moreover, we present a knowledge-enhanced HSI classification network comprising several novel knowledge-calibrated cross-Mamba (KCCM) modules. These modules can seamlessly inject the cross-scene transferable semantic knowledge and the long-range spatial dependency into feature representation, thus producing better generalization capacity in unseen target HSI even with scarce pixel-wise annotations. Sufficient experimental results verify the efficacy of the proposed method in terms of CSFS-HSIC. The code will be available at https://github.com/zhaoxb2025/PSBL-main**

***Index Terms*—Few-shot learning (FSL), hyperspectral image (HSI) classification, Mamba.**

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I. Introduction

HYPERSPECTRAL imaging is a powerful remote sensing technology that concurrently captures the spatial and spectral information of surface features, generating a 3-D data cube, termed a hyperspectral image (HSI) [1]. In contrast to conventional RGB or panchromatic images, each pixel in an HSI carries a high-dimensional spectral curve. This curve acts as an "optical fingerprint," enabling fine-grained discrimination of materials. This unique attribute has driven the widespread application of HSIs across diverse fields, such as environmental monitoring [2], [3], precision agriculture [4], and geological exploration. As a fundamental task underpinning many such applications, HSI classification (HSIC) seeks to assign a semantic label to each pixel within an image [5]. Driven by the great success of deep learning in the computer vision domain, numerous deep neural networks have been developed for HSIC [6], delivering promising performance [7], [8]. This success hinges on substantial pixel-wise annotations for network training. In real remote sensing scenarios, collecting sufficient pixel-wise annotations is often infeasible due to high annotation costs. Thus, increasing attention is being directed to few-shot HSI classification (FS-HSIC), which aims to assign pixel-wise semantic labels to target HSIs with only limited such annotations [9].

Existing FS-HSIC approaches predominantly adhere to two methodological paradigms: single-domain FS-HSIC [10], [11] and domain adaptation-based FS-HSIC [12], [13], [14]. The first paradigm aims to exploit the sparse semantic information inherent in the limited annotated pixels of a target HSI by employing specialized network architectures and loss functions [15]. However, due to the fundamental constraint of extremely scarce supervision during training, the resulting models often exhibit limited generalization capability—particularly in challenging scenarios characterized by significant intraclass variation and subtle interclass distinctions. Conversely, the second paradigm seeks to enhance few-shot learning (FSL) by transferring additional semantic knowledge from a source domain containing extensive pixel-wise annotations for categories identical to those in the target HSI [12]. Although these methods generally achieve stronger generalization performance, their practical implementation faces a critical limitation: the acquisition of a suitable source domain with perfectly aligned categories is often infeasible in

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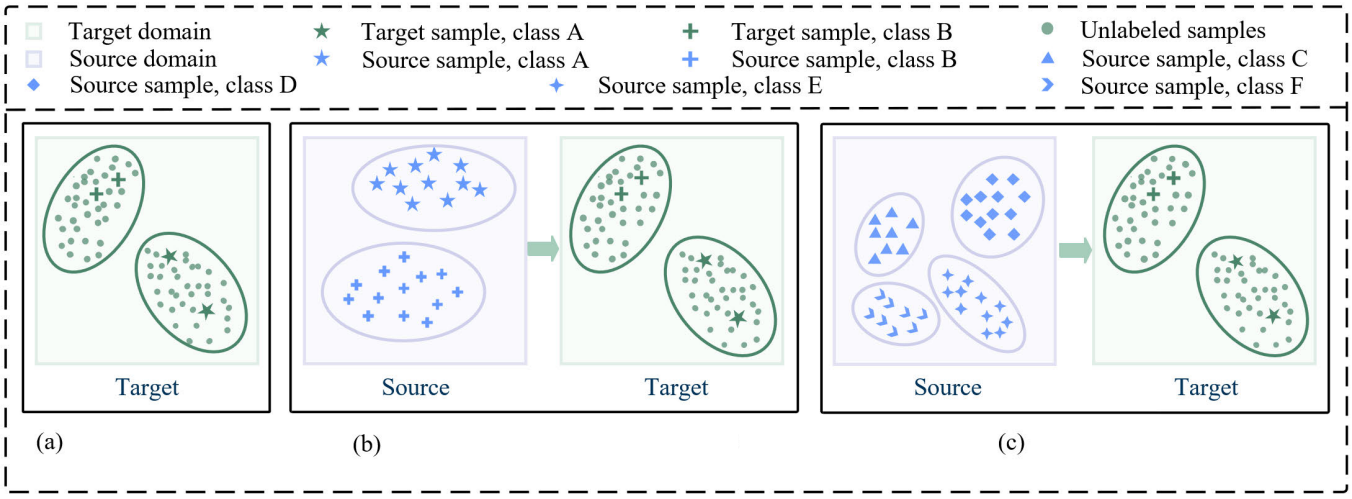


Fig. 1. Comparison of the CSFS-HSIC with existing paradigms. (a) Single-domain few-shot HSI classification (FS-HSIC). This paradigm can only explore sparse spatial–spectral knowledge within the target image. (b) Domain adaptation-based FS-HSIC. This paradigm transfers semantic knowledge from the source domain to the target domain, yet requires identical categories between the source and target domains. (c) CSFS-HSIC. This paradigm enables the transfer of semantic knowledge to the target image across scenes containing different categories.

real-world scenarios [16]. In contrast, it is feasible to collect extensive annotated images with categories different from those of the target HSIs, referred to as cross-scene images in this study. However, existing methods rarely leverage transferable cross-scene semantic knowledge to enhance FSL for the target HSIs.

To address this gap, we first formalize the cross-scene few-shot HSI classification (CSFS-HSIC) task that seeks to exploit cross-scene transferable semantic knowledge from an auxiliary dataset with sufficient pixel-wise annotations to regularize the FSL on target HSIs of different categories. A comparison of the three paradigms can be found in Fig. 1. Under this setting, we propose a novel spectral basis learning-based CSFS-HSIC method (PSBL), comprising a prototypical spectral basis learning network (PSBLN) and a knowledge-enhanced HSI classification network (KECHN). Specifically, the PSBLN takes prototypical spectra (e.g., centroids) of all categories in the auxiliary dataset as input to generate spectral bases. These bases, when exploited in an auto-encoding paradigm augmented with an additional classification loss, are not only discriminative but also enable accurate linear reconstruction of prototypical spectra for all categories—allowing the network to distill cross-scene transferable semantic knowledge in the spectral domain. Meanwhile, the KECHN is developed by stacking novel knowledge-calibrated cross-Mamba (KCCM) modules. These modules adapt prevalent cross-attention and Mamba modules to seamlessly inject hierarchical cross-scene transferable semantic knowledge and long-range spatial dependencies into feature representations. In this way, the proposed method yields stronger generalization ability on unseen target HSIs, even with scarce pixel-wise annotations. In experiments, we establish a standard evaluation benchmark and clarify the effectiveness of the proposed method in terms of CSFS-HSIC, providing a promising solution to this challenging yet practically significant problem.

The main contributions of this work can be summarized as follows.

1) We take the first attempt to formalize the CSFS-HSIC task and propose a novel PSBL method that can appropriately transfer the cross-scene semantic knowledge into the FSL on the target HSI with better generalization capacity.
2) We design a PSBLN that extracts hierarchical and cross-scene transferable spectral semantic knowledge from an auxiliary annotated dataset.
3) We develop a KCCM Module that seamlessly injects cross-scene transferable semantic knowledge and long-range spatial dependencies into feature representations, thereby enhancing generalization in unseen target scenes.
4) We establish a standard CSFS-HSIC evaluation benchmark and report the new state-of-the-art performance on it.

The remainder of this article is organized as follows. Section II reviews related work. Section III elaborates on the proposed PSBL methodology. Section IV presents experimental results and analysis. Finally, Section V concludes the article and discusses future research directions.

II. Related Work

This section begins with a comprehensive review of FS-HSIC and its representative studies. It then elucidates the attention mechanisms relevant to our work.

A. Few-Shot HSI Classification

FS-HSIC primarily consists of two categories: single-domain FS-HSIC [17], [18], and domain adaptation-based FS-HSIC [12]. Single-domain approaches focus on developing sophisticated model architectures and loss functions to maximize knowledge extraction from limited annotations within the target domain [19]. Meta learning frameworks have been particularly influential in this area. For instance, prototypical networks [17] and relation network [11] have

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been applied to FS-HSIC by learning metric spaces where classification is performed by computing distances to class prototype representations [20]. Data augmentation techniques have also been utilized to alleviate data scarcity [21]. Generative adversarial networks (GANs) have been employed to create realistic synthetic HSI patches, thereby expanding the effective training set size [22], [23]. In addition, self-supervised [17], [24], and semi-supervised [25] learning are studied to mitigate the problem of insufficient labeled data. Despite these innovations, single-domain methods face fundamental limitations: their performance is inherently constrained by the extreme scarcity of supervisory signals. In complex scenes with significant intraclass variation and subtle interclass differences, limited annotated samples often fail to capture the true data distribution, leading to suboptimal generalization. In addition, these methods typically require careful regularization to prevent overfitting to small training sets, potentially limiting their capacity to learn discriminative features.

Domain adaptation methods attempt to overcome the limitations of single-domain approaches by transferring knowledge from source domains with abundant annotations to target domains with limited labels [26], [27]. These methods typically operate under the assumption that, despite distribution shifts between domains, invariant features exist that can be leveraged for effective knowledge transfer [28]. Adversarial domain adaptation has shown considerable promise in FS-HSIC [27]. Liu et al. [29] proposed an adversarial domain adaptation framework that aligns feature distributions between source and target domains using a domain discriminator. To further enhance discrimination capability, Liu et al. [27] introduced a multiscale conditional domain discriminator to mitigate domain shift. Recently, research focus has shifted toward multisource domain adaptation, which leverages multiple source domains to enhance adaptation performance [30]. Liang et al. [31] introduced a multisource domain adaptation method that dynamically weights contributions from different source domains based on their relevance to the target domain, demonstrating enhanced robustness to domain shift compared to single-source methods. While these methods partially alleviate the data scarcity issue and enhance model generalization, they rely on the strict assumption of category alignment between source and target domains—a condition rarely satisfied in real-world scenarios [16]. Crucially, collecting large-scale multicategory cross-scene images is more practical than satisfying this stringent requirement. To address this gap, we formalize the CSFS-HSIC task and subsequently propose a novel PSBL method. Without requiring category-aligned source domains, the proposed method achieves robust few-shot performance, effectively overcoming the dual limitations of both single-domain and domain adaptation-based FS-HSIC methods.

B. Attention Mechanism

Contemporary attention-based models are dominated by the Transformer architecture [32] and the recently introduced Mamba architecture [33]. The Transformer, built upon the self-attention mechanism, fundamentally revolutionized sequence modeling by eliminating recurrent structures [34]. Its core operation is the self-attention, formulated as

$$\text{Attention}\,(Q,K,V)=\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where Q, K, and V represent the query, key, and value matrices, respectively, and d_k denotes the feature dimension. To enhance representational capacity, the Transformer employs a multihead attention mechanism, defined as

$$\text{MultiHead}\,(Q,K,V)=\text{Concat}\,(\text{head}_1,\ldots,\text{head}_h)\,W^O \tag{2}$$

where each attention head is computed as

$$\text{head}_i=\text{Attention}\left(QW_i^Q,K\ W_i^K,V\ W_i^V\right). \tag{3}$$

The Transformer offers three principal advantages: 1) parallel token-level computation; 2) powerful long-range dependency modeling capabilities; and 3) enhanced interpretability through attention weights. Despite these strengths, its primary limitation lies in the quadratic computational complexity $\mathcal{O}(n^2)$ with respect to sequence length n, resulting in substantial memory and computational overhead. Furthermore, Transformer models typically require extensive training datasets, constraining their performance in FSL scenarios.

To address the limitation of quadratic complexity, researchers have developed Mamba—a sequence modeling framework based on state space models that achieves linear computational complexity. Its theoretical foundation originates from the continuous-time system representation

$$\frac{dh\,(t)}{dt}=Ah\,(t)+Bx\,(t),\quad y\,(t)=Ch\,(t) \tag{4}$$

where $h(t)$ denotes the hidden state, $x(t)$ denotes the input, and $y(t)$ denotes the output. Using zero-order hold discretization with a timescale parameter Δ, the continuous system is discretized as

$$h_t=\bar{A}h_{t-1}+\bar{B}x_t,\quad y_t=Ch_t \tag{5}$$

where $\bar{A}=\exp(\Delta A)$ and $\bar{B}=(\Delta A)^{-1}(\exp(\Delta A)-I)\cdot\Delta B$ represent the discretized state transition and input matrices, respectively.

The core architectural innovation of Mamba is the S6 model (structured state space sequence model), which introduces a selective mechanism. Unlike traditional state space models where parameters B, C, and Δ remain static, S6 generates these parameters dynamically as functions of the input x_t

$$B_t=f_B\,(x_t),\quad C_t=f_C\,(x_t),\quad \Delta_t=f_\Delta\,(x_t). \tag{6}$$

This input-dependent parameterization enables S6 to actively select relevant information from sequences, effectively implementing a content-aware reasoning mechanism. The selective mechanism allows S6 to approximate the contextual processing capabilities of attention-based models while maintaining the computational efficiency of recurrent neural networks. Although Mamba significantly reduces computational complexity, conventional vision Mamba [35] exhibits limited generalization in FS-HSIC due to insufficient annotated data. To enhance model generalization, we integrate cross-attention with an improved Mamba architecture, proposing a novel KCCM module. This innovative design enables seamless

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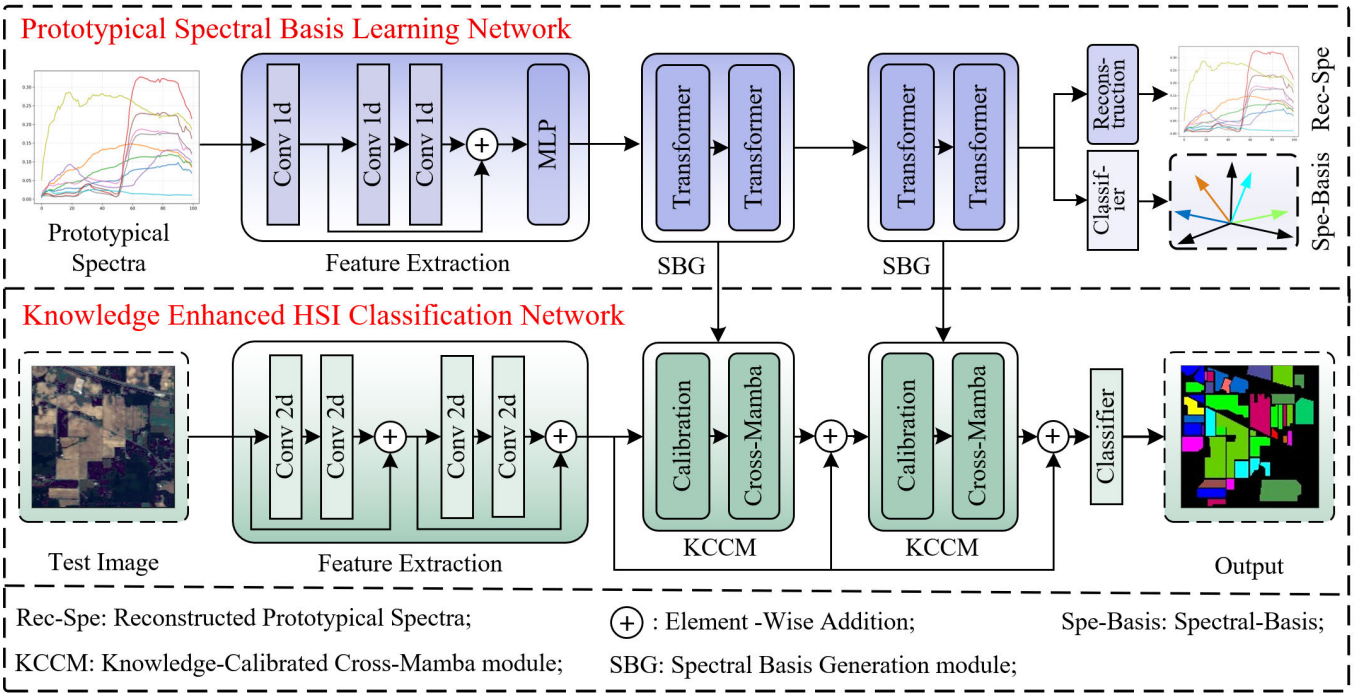


Fig. 2. Overall architecture of the proposed PSBL. This architecture comprises two core components: PSBLN and KECHN. PSBLN extracts cross-scene transferable semantic knowledge from auxiliary datasets. KECHN integrates multiple KCCM modules to perform knowledge-enhanced HSI classification. Each KCCM module incorporates a calibration submodule and a cross-Mamba submodule, enabling seamless injection of cross-scene transferable semantic knowledge and long-range dependencies into feature representations to produce superior generalization capabilities.

integration of transferable semantic knowledge across scenes and long-range spatial dependencies into feature representations, significantly enhancing generalization capacity.

III. Proposed Method

In this section, we first formalize the CSFS-HSIC task. The overall architecture of the PSBL framework is subsequently presented, followed by a detailed description of its two subnetworks: PSBLN and the KECHN.

A. CSFS-HSIC Definition

The CSFS-HSIC has emerged as a critical and challenging research direction in remote sensing. As shown in Fig. 1(c), this task aims to transfer knowledge from a source domain $\mathcal{D}_s = \{(X_i^s, Y_i^s)\}_{i=1}^{N_s}$ with abundant labeled data to a target domain $\mathcal{D}_t = \{(X_j^t, Y_j^t)\}_{j=1}^{N_t}$ with different categories, thereby enabling accurate ground objects classification under few-shot setting. Here, X_i^s and X_j^t denote HSI from the source domain and target domain, respectively; Y_i^s and Y_j^t represent corresponding semantic labels; N_s and N_t denote the number of labeled samples in each domain ($N_t \ll N_s$); and $\mathcal{C}_s$ and $\mathcal{C}_t$ represent their respective category sets ($\mathcal{C}_t \nsubseteq \mathcal{C}_s$). Formally, this task aims to learn a generalized classification model $f : \mathcal{X} \to \mathcal{Y}$ to minimize the expected loss $\mathbb{E}_{(X,Y)\sim\mathcal{D}_t}[\mathcal{L}(f(X), Y)]$. Since the target domain has different categories from the source domain, one challenge of CSFS-HSIC lies in the semantic gap. Furthermore, due to the lack of labeled data in the target domain, CSFS-HSIC frequently suffers from overfitting. Therefore, bridging the semantic gap to achieve effective knowledge transfer on unseen target classes and mitigating overfitting are crucial for addressing this problem.

B. Overall Framework

To address the CSFS-HSIC task, we propose a novel PSBL approach, which aims to enhance the model's generalization capability on the target HSI with scarce annotations by leveraging cross-scene transferable semantic knowledge of the $\mathcal{D}_s$. As illustrated in Fig. 2, the proposed method comprises two subnetworks: PSBLN and KECHN. PSBLN learns transferable semantic knowledge from $\mathcal{D}_s$. This model employs an autoencoder paradigm with enhanced classification constraints to extract hierarchical spectral bases from representative spectra across all categories in the $\mathcal{D}_s$, enabling discriminative feature learning. KECHN performs knowledge-enhanced input HSI classification. It incorporates multiple KCCM modules that seamlessly inject cross-domain transferable semantic knowledge and long-range spatial dependencies into feature representations through integrated cross-attention and cross-Mamba, thereby enhancing generalization on target HSI with scarce annotations.

C. Prototypical Spectral Basis Learning Network

PSBLN aims to learn cross-scene transferable semantic knowledge in the spectral domain by learning spectral bases from $\mathcal{D}_s$. As shown in Fig. 2, the network comprises two interconnected components: feature extraction and spectral bases generation, responsible for initial feature extraction and hierarchical spectral bases generation, respectively.

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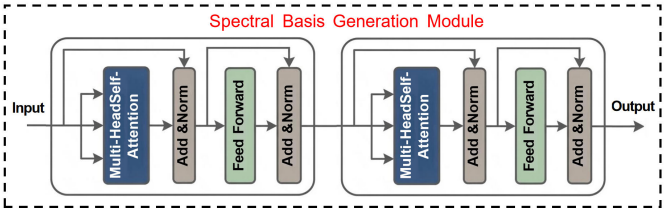


Fig. 3. Schematic of the SBG module. It comprises two cascaded Transformer layers, each with four attention heads and a dropout rate of 0.1.

1) Feature Extraction: The input to PSBLN consists of prototypical spectra for all categories in the $\mathcal{D}_s$. For every class $c \in \mathcal{C}_s$, we compute the centroid by averaging randomly selected pixel spectra (up to 200 samples per class). This yields the input $X_S \in \mathbb{R}^{|\mathcal{C}_s| \times D}$, where $|\mathcal{C}_s|$ denotes the number of classes, and D represents the spectral dimension. The network first extracts initial features from the input through a 1-D convolutional layer, followed by a residual block:

$$F_p = \text{Res1D}\left(\text{Conv1D}\left(X_S\right)\right) \tag{7}$$

where Conv1D(·) and Res1D(·) denote the convolutional and residual operations, respectively.

Since the $X_S \in \mathbb{R}^{|\mathcal{C}_s| \times D}$ may contain redundant categories, we employ a single-layer multilayer perceptron (MLP) to compress it into M distinct compact representations, thereby capturing the most essential spectral representations. This is formalized in the following equation:

$$Z_p = \sigma\left(W_p F_p + b_p\right) \tag{8}$$

where $W_p \in \mathbb{R}^{M \times |\mathcal{C}_s|}$ and $b_p \in \mathbb{R}^M$ are learnable parameters, and $\sigma(\cdot)$ is the ReLU activation function.

2) Spectral Bases Generation: The primary objective of PSBLN is to generate hierarchical spectral bases that encapsulate transferable semantic knowledge. This is achieved through two cascaded spectral basis generation (SBG) modules. As depicted in Fig. 3, each SBG module consists of two sequentially connected Transformer layers. Each Transformer layer includes a multihead self-attention (with four attention heads; dropout = 0.1) and a feedforward neural network (FFN). These two key submodules capture discriminative spectral features and enhance the model's expressive power through nonlinear transformations, respectively. The generation process of the lth spectral basis can be formalized as

$$P^{(l)} = \text{SBG}_l\left(P^{(l-1)}\right), \quad l = 1, 2 \tag{9}$$

where $P^{(0)} = Z_p$ and SBG_l represents the lth SBG module. The outputs $P^{(1)}$ and $P^{(2)}$ represent the generated hierarchical spectral bases.

Next, a reconstruction head consisting of a single-layer MLP is employed to linearly reconstruct the original prototypical spectra from $P^{(2)}$, and a classification head containing a fully connected layer is used to classify the spectral basis vectors of $P^{(2)}$.

3) Loss Function: To train the PSBLN, we impose both a discriminative loss and a reconstructive loss, which, respectively, encourage the diversity and transferability of the generated spectral basis vectors. Specifically, the discrimination loss is formulated as a cross-entropy objective over the basis vectors derived from $P^{(2)}$, enforcing their distinctiveness and nonredundancy, thereby preserving class-specific spectral characteristics

$$\mathcal{L}_{\text{CE}}^{\text{PB}} = -\frac{1}{M}\sum_{i=1}^{M}\sum_{c=1}^{M} g_{i,c} \log \hat{g}_{i,c} \tag{10}$$

where M denotes the number of basis vectors, $g_{i,c}$ is the ground-truth (GT) label, and $\hat{g}_{i,c}$ represents the predicted probability for class c.

To further promote generalization, a reconstructive loss is introduced based on the ℓ_1 distance between the reconstructed and input prototypical spectra. This constraint ensures that the network can faithfully recover the original spectral prototypes from the learned basis, thereby capturing class-agnostic shared representations

$$\mathcal{L}_{\text{REC}}^{\text{PB}} = \|X_S - \hat{X}_S\|_1 \tag{11}$$

where $\hat{X}_S$ denotes the reconstructed prototypical spectra.

By jointly optimizing the discriminative and reconstructive objectives, the learned spectral basis vectors are encouraged to be both discriminative and transferable, enabling the distillation of cross-scene semantic knowledge from diverse spectral domains into a compact and shared spectral basis representation.

D. Knowledge-Enhanced HSI Classification Network

The KECHN is designed to achieve accurate HSIC by injecting transferable spectral knowledge and long-range spatial dependencies. As illustrated in Fig. 2, the network architecture comprises a feature extraction module followed by two KCCM modules. The pipeline begins with feature extraction, followed by refinement through cascaded KCCM modules—which significantly enhance the model's generalization capability by injecting cross-scene transferable semantic knowledge and long-range spatial dependencies.

1) Knowledge-Calibrated Cross-Mamba: The KCCM module is designed to seamlessly inject cross-scene transferable semantic knowledge and long-range dependencies into feature representations. Each KCCM comprises two core components: the calibration submodule integrates cross-scene transferable semantic knowledge through cross-attention, while the cross-Mamba submodule captures long-range spatial dependency via an efficient cross-axial scanning mechanism.

a) Calibration: The calibration submodule employs a cross-attention mechanism to integrate cross-scene transferable semantic knowledge into the input feature representation while mitigating spectral bias. As illustrated in Fig. 4, for an input feature map $X \in \mathbb{R}^{H \times W \times C}$, the module first refines the representation through two 2-D convolutional layers

$$X_1 = \text{Flatten}\left(\text{Conv2D}_2\left(\text{Conv2D}_1\left(X\right)\right)\right) \tag{12}$$

where Flatten(·) reshapes the spatial dimensions to a sequence of length $N = H \times W$, yielding $X_1 \in \mathbb{R}^{N \times C}$.

The spectral basis $P^{(l)} \in \mathbb{R}^{M \times D}$ from PSBLN are projected to match the feature dimension C

$$P_p^{(l)} = P^{(l)} W_p, \quad W_p \in \mathbb{R}^{D \times C} \tag{13}$$

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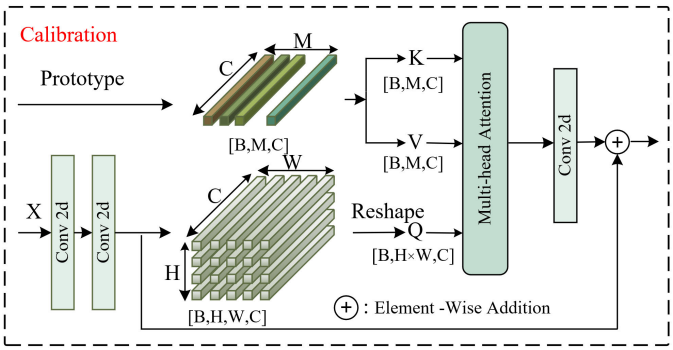


Fig. 4. Overview of the calibration submodule. The generated spectral bases serve as keys and values, while HSI features act as queries. A multihead attention mechanism integrates these components to calibrate spectral bias and integrate semantic knowledge. The resulting representation is refined through 1×1 convolutional layers with residual connections, ultimately enhancing feature representations.

where $P_p^{(l)} \in \mathbb{R}^{M\times C}$ represents the dimension-aligned spectral bases.

The cross-attention operation is then applied

$$\hat{X} = \text{CrossAttn}\left(X_1, P_p^{(l)}\right) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V \tag{14}$$

where $Q = X_1 W_Q$, $K = PW_K$, and $V = PW_V$ are query, key, and value projections, with learnable matrices $W_Q, W_K, W_V \in \mathbb{R}^{C\times d}$, respectively. The $\sqrt{d}$ is a scaling factor that ensures stable gradients.

The resulting representations are refined through a 1×1 convolutional layer with residual connections, thereby enhancing the feature representation capabilities

$$\tilde{X} = \text{Conv}_{1\times 1}\left(\hat{X}\right) + X. \tag{15}$$

where $\text{Conv}_{1\times 1}(\cdot)$ denotes a 2-D 1×1 convolutional layer.

The calibration module effectively facilitates knowledge transfer while mitigating spectral bias.

b) Cross-Mamba: The cross-Mamba submodule is designed to capture long-range dependencies within input HSIs. While some existing studies have attempted to model such dependencies using Transformer architectures, the high parameter count of Transformers often leads to limited generalization capability in few-shot HSIC, along with high computational complexity. To address the generalization constraints associated with Transformers, alternative approaches have explored the use of Mamba—a model with lower parameter requirements and computational overhead—for effectively capturing long-range dependencies in the input image. Existing Mamba-based HSIC methods typically partition the input HSI into overlapping image blocks, then perform spatial modeling through multidirectional scanning [8], [36], [37]. While this avoids increased computational complexity from overly long sequences, it limits the model to exploiting only local spatial dependencies, resulting in limited generalization capabilities in FSL scenarios.

In contrast to existing approaches, the cross-Mamba submodule captures global contextual spatial dependencies in a patch-free manner, demonstrating robust generalization capabilities. Specifically, cross-Mamba employs an efficient cross-axial scanning approach to capture pixel-level long-range dependencies, achieving more comprehensive dependency modeling while reducing computational complexity. As illustrated in Fig. 5, this method transforms 2-D spatial modeling into parallel unidirectional sequence processing along horizontal and vertical dimensions, with each sequence incorporating positional embeddings to preserve spatial location information. After integrating positional embeddings, a shared S6 model evolves the state space to model horizontal and vertical spatial dependencies, followed by bidirectional integration to capture global spatial dependencies. This design reduces the required sequence length for pixel-level global spatial modeling from $H \times W$ to $\max(H, W)$, while enabling cross-dimensional parameter sharing across sequence-space models, significantly reducing the number of parameters.

Specifically, for an input $X \in \mathbb{R}^{B\times H\times W\times C}$ (where B denotes the batch size), we unfold it along the horizontal and vertical directions, respectively, and fold the spatial dimensions into the batch dimension to obtain horizontal sequence $X_h \in \mathbb{R}^{(B\times H)\times W\times C}$ and vertical sequence $X_w \in \mathbb{R}^{(B\times W)\times H\times C}$. This reduces the original pixel-wise sequence length from $H \times W$ to $\max(H,W)$.

Since the unfolding operation inevitably disrupts spatial information—which is critical for spatial modeling—we introduce learnable dynamic positional embeddings to preserve positional information

$$X_{ph} = X_h + P_h(\theta), \quad X_{pw} = X_w + P_w(\theta) \tag{16}$$

where $P_h(\theta)$ and $P_w(\theta)$ are learnable parameters.

Next, the sequences with position embeddings undergo state space evolution via a shared S6 model

$$\begin{aligned} s_t &= As_{t-1} + Bu_t, \quad s_0 = 0 \\ o_t &= C_s s_t + Du_t. \end{aligned} \tag{17}$$

Subsequently, the sequence output by the S6 model is projected back to the original spatial dimension, yielding Z_{ph} and Z_{pw}, respectively. Then, bidirectional integration is performed via concatenation and 1×1 convolution to capture long-range spatial dependencies

$$Z = \text{Conv}_{1\times 1}\left(\text{Concat}\left[Z_{ph}, Z_{pw}\right]\right). \tag{18}$$

The final output is obtained via a residual connection

$$\tilde{X} = Z + X. \tag{19}$$

This cross-scan pipeline enables efficient pixel-wise dependency capture while maintaining low parameterization through weight sharing and optimized sequence processing.

2) KCCM Integration in KECHN: Within the KECHN, feature extraction is first performed on the input HSI $X \in \mathbb{R}^{H\times W\times C}$ via two sequential residual blocks

$$F_0 = \text{Res2D}_2\left(\text{Res2D}_1\left(X\right)\right) \tag{20}$$

where $\text{Res2D}_i(\cdot)$ denotes a two-layer 2-D residual convolutional block.

The extracted feature F_0 is progressively refined via two cascaded KCCM modules, with the input to the second module incorporating the residual F_0

$$F = \text{KCCM}_2\left(\text{KCCM}_1\left(F_0\right) + F_0\right). \tag{21}$$

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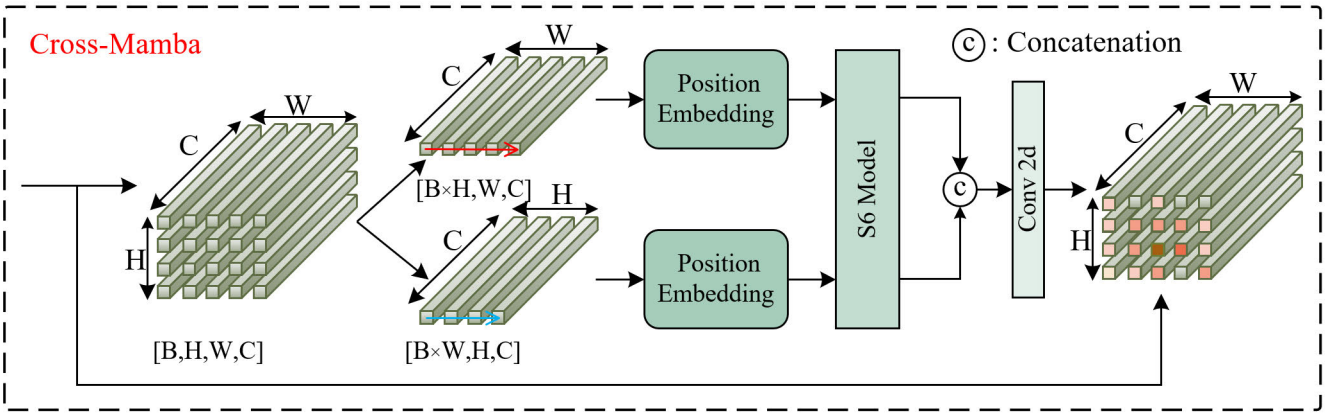


Fig. 5. Overview of the cross-Mamba submodule. Input features are unfolded along horizontal and vertical directions to generate oriented sequences. Each sequence is augmented with positional embeddings and captures spatial dependencies through a shared S6 model. The resulting representations are concatenated, refined via 1×1 convolutions, and fused through residual connections to produce enhanced spatial features.

Each KCCM module seamlessly integrates cross-scene transferable semantic knowledge and long-range dependencies, significantly enhancing the model's generalization capability.

Finally, the refined features F are fused with F_0 and predicted through a classifier containing a single 1×1 convolutional layer, expressed as

$$\hat{Y} = \text{Conv}_{1\times1}(F + F_0) \tag{22}$$

where $Y \in \mathbb{R}^{H\times W\times \#class}$ denotes the predict result and $\#class$ is number of class.

3) Loss Function: The KECHN is optimized using a pixel-wise cross-entropy loss function

$$\mathcal{L}_{\text{CE}}^{\text{IMG}} = -\frac{1}{N'}\sum_{i=1}^{N}\text{mask}_i \sum_{c=1}^{\#class} y_{i,c}\log \hat{y}_{i,c} \tag{23}$$

where $y_{i,c}$ and $\hat{y}_{i,c}$ represent the GT and predicted probability for class c at pixel i, respectively. mask_i indicates annotated pixel positions. $N' = \sum_{i=1}^{N}\text{mask}_i$ denotes the total number of labeled samples. N represents the total number of pixels.

E. Training and Inference Protocol

The proposed PSBL comprises two procedures: training and inference.

1) Training: In the training, the proposed PSBL is first pretrained on the $\mathcal{D}_s$, followed by fine-tuning using a few labeled samples from the target images. Different loss functions are employed at each stage. In the pretraining stage, PSBLN optimizes a composite loss function by combining the discriminative loss $\mathcal{L}_{\text{CE}}^{\text{PB}}$ with the reconstructive loss $\mathcal{L}_{\text{REC}}^{\text{PB}}$. In contrast, KECHN is trained using the image classification loss function $\mathcal{L}_{\text{CE}}^{\text{IMG}}$. The overall pretraining loss function is defined as (24). In the fine-tuning stage, we first randomly initialize a network with an identical structure to the pretrained backbone, and adjust the number of classes in its KECHN to match that of the target image. Subsequently, the parameters of the PSBLN in the pretrained backbone are transferred to the new network and kept frozen, so as to fully exploit the cross-scene transferable semantic knowledge acquired during pretraining. For the pretrained KECHN, only the first four layers are transferred to the new network and kept trainable, to leverage the more general shallow-level features. Optimization in this stage relies solely on the image classification $\mathcal{L}_{\text{CE}}^{\text{IMG}}$

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{CE}}^{\text{IMG}} + \mathcal{L}_{\text{REC}}^{\text{PB}} + \mathcal{L}_{\text{CE}}^{\text{PB}}. \tag{24}$$

2) Inference: In the inference, unlabeled samples in the target images are predicted using the fine-tuned model, and the results are evaluated.

IV. Experiments

In this section, the dataset construction procedures are first introduced. Subsequently, the experimental details and configurations are described, including the comparison methods, parameter settings, and evaluation metrics. Finally, a comprehensive comparative analysis of the experimental results is presented. All experiments were performed on an Intel[1] Xeon[1] Platinum 8350C CPU at 2.60 GHz, a GeForce RTX 3090 (24 GB), and a Ubuntu 20.04 system, and the open-source software framework is PyTorch.

A. Dataset Construction

Existing HSI datasets exhibit significant heterogeneity in spectral characteristics—including spectral range, resolution, number of bands, and spatial dimensions—which hinders the integration of multisource data for training unified models. To enable unified CSFS-HSIC model training using these datasets, we propose a spectral standardization scheme to construct a standardized evaluation benchmark. This scheme comprises two core steps: wavelength range selection and band number alignment.

Formally, let a collection of raw datasets be denoted as $\mathcal{O} = \{\mathcal{O}_1, \mathcal{O}_2, \ldots, \mathcal{O}_n\}$, where each dataset $\mathcal{O}_i = \{X_i, Y_i\}$ consists of an HSI $X_i \in \mathbb{R}^{B_i\times H_i\times W_i}$ (with B_i, H_i, and W_i representing the number of bands, height, and width, respectively) and its corresponding label map Y_i. The number of classes in each dataset is given by $|Y_i|$.

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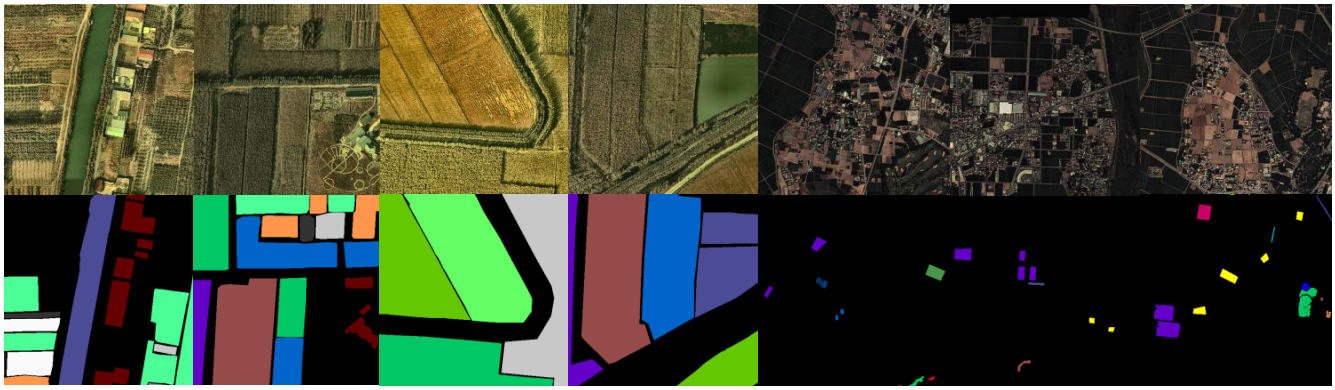


Fig. 6. Representative samples of the training set. (Top row) Pseudo-color composites. (Bottom row) Corresponding GT.

1) Wavelength Range Selection: For each HSI X_i, spectral bands within the 400–1000-nm range are extracted as

$$X_i' = \{x_{i,\lambda} \mid \lambda \in [400, 1000]\}$$

where $x_{i,\lambda}$ denotes the spectral band corresponding to wavelength λ.

2) Band Number Alignment: Each processed image X_i' is normalized to a unified band number: images with fewer than 100 bands are expanded to 100 via interpolation, while those exceeding 100 bands are reduced to 100 through uniform sampling. This yields a consistent 100-band representation across all datasets.

After this two-stage standardization, we obtain a spectrally unified dataset collection $\mathcal{O}' = \{\mathcal{O}_1', \mathcal{O}_2', \ldots, \mathcal{O}_n'\}$. These datasets are partitioned into a training set (source domain) $\mathcal{D}_s$ and a test set (target domain) $\mathcal{D}_t$. For $\mathcal{D}_s$, a unified label space covering all categories is formed by reinitializing and merging the label sets. Subsequently, all image-label pairs are padded and cropped to 512×512 to support batch processing. For $\mathcal{D}_t$, all image-label pairs are kept unchanged to preserve original spatial relationships, enabling the modeling of long-range dependencies.

Following the above procedures, the resulting training and test sets are obtained.

3) Training Set: The training set incorporates eleven publicly hyperspectral datasets: Washington DC Mall, XiongAn, XuZhou, HongHu, LongKou, Shanghai, Botswana, Chikusei, Pavia University, Berlin, and Houston2018. Following label space unification procedures, the training set retains 139 semantically distinct ground object categories encompassing diverse environmental scenarios and material types. Information regarding their sensor types, ground sampling distance (GSD), and number of classes is presented in Table I. Representative visualizations of selected source domain samples are presented in Fig. 6.

4) Test Set: The test set incorporates five benchmark datasets: Indian Pines (IP), Salinas (SA), Loukia (LK), HanChuan (HC), and Kennedy Space Center (KSC). These datasets were selected based on their sensor diversity, spatial resolution variations, and representative ground objects characteristics. Key attributes, including sensor types, spatial resolutions, GT, and class distributions, are summarized in Table II.

TABLE I
DETAILS OF THE RAW HSI DATASET IN THE TRAINING SET

Datasets	Sensors	GSD	#Classes
Washington DC Mall	Hydice	3 m	7
XiongAn	GF-5	0.5 m	19
XuZhou	HySpex SWIR	0.73 m	9
HongHu	Headwall Hyperspec	0.463 m	22
LongKou	Headwall Hyperspec	0.043 m	9
Shanghai	EO-1	30 m	3
Botswana	EO-1	30 m	14
Chikusei	Headwall Hyperspec	1.3 m	19
Pavia University	ROSIS	1.3 m	9
Berlin	HyMap	13.8 m	8
Houston2018	-	0.5 m	20

B. Experiment Setting

1) Comparison Methods: We evaluate our approach against four categories of baselines: the classical SVM [38]; state-of-the-art HSIC methods, including SSTN [TGRS2022] [39], A^2S^2K-ResNet [TGRS2022] [40], and SSSAN [TGRS2021] [41]; single-domain FS-HSIC methods such as DM-MRN [TGRS2024] [11] and SSEFN [TGRS2025] [10]; and domain adaptation-based FS-HSIC methods, including HyMuT [TGRS2024] [14], FDFSL [TGRS2024] [42], CDMLC [TGRS2025] [43], and MEDPL [TGRS2025] [44].

2) Sample Selection: In the source domain, 200 labeled samples were randomly selected from each class to pretrain the model. For classes with fewer than 200 samples, 80% of the available instances were randomly chosen instead. In the target domain, unless otherwise specified, five labeled samples per class were randomly selected for model fine-tuning and subsequent evaluation.

3) Optimization: The detailed hyperparameter settings are summarized in Table III. Both the pretraining and fine-tuning phases employed the Adam optimizer, with the learning rate

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TABLE II
DETAILS OF FIVE TARGET DATASETS

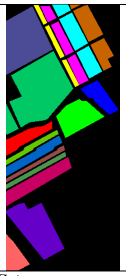
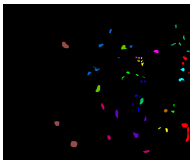
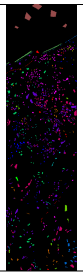
Datasets		IP		SA		HC		KSC		LK	
Sensors		AVIRIS		AVIRIS		Headwall Nano-Hyperspec		AVIRIS		HyMap	
Size		145 × 145		512 × 217		1217 × 303		512 × 614		945 × 249	
GSD		20 m		3.7 m		0.109 m		18 m		30 m	
											
No.	Color	Category	#Smp.	Category	#Smp.	Category	#Smp.	Category	#Smp.	Category	#Smp.
1		Alfalfa	46	Brocoli_green_weeds_1	2009	Strawberry	6710	Scrub	761	Dense urban fabric	288
2		Corn-notill	1428	Brocoli_green_weeds_2	3726	Cowpea	3412	Willow swamp	243	Mineral extraction sites	67
3		Corn-mintill	830	Fallow	1976	Soybean	1543	CP hammock	256	Non-irrigated arable land	542
4		Corn	237	Fallow_rough_plow	1394	Sorghum	802	Slash pine	252	Fruit trees	79
5		Grass-pasture	483	Fallow_smooth	2678	Water spinach	180	Oak/Broadleaf	161	Olive groves	1401
6		Grass-tree	730	Stubble	3959	Watermelon	679	Hardwood	229	Broad-leaved forest	223
7		Grass-pasture-mowed	28	Celery	3579	Greens	885	Swap	105	Coniferous forest	500
8		Hay-windrowed	478	Grapes_untrained	11271	Trees	2696	Graminoid marsh	431	Mixed forest	1072
9		Oats	20	Soil_vinyard_develop	6203	Grass	1420	Spartina marsh	520	Dense sclerophyllous vegetation	3793
10		Soybean-notill	972	Corn_senesced_green_weeds	3278	Red roof	1577	Cattail marsh	404	Sparce sclerophyllous vegetation	2803
11		Soybean-mintill	2455	Lettuce_romaine_4wk	1068	Gray roof	2536	Salt marsh	419	Sparsely vegetated areas	404
12		Soybean-clean	593	Lettuce_romaine_5wk	1927	Plastic	551	Mud flat	503	Rocks and sand	487
13		Wheat	205	Lettuce_romaine_6wk	916	Bare soil	1367	Water	927	Water	1393
14		Woods	1265	Lettuce_romaine_7wk	1070	Road	2784			Coastal water	451
15		Buildings-Grass-Trees-Drives	386	Vinyard_untrained	7268	Bright object	170				
16		Stone-Steel-Towers	93	Vinyard_vertical_trelis	1807	Water	11310				
Total			10249		54129		38622		5211		13503

TABLE III
OPTIMIZATION CONFIGURATIONS FOR PRETRAINING AND FINE-TUNING

Hyperparameter	Pretraining	Fine-tuning
Batch size (b)	2	1
Epoch (E)	2000	400
Learning rate	0.0005	0.0005
Weight decay	0.0002	0.0002

and weight decay set to 0.0005 and 0.0002, respectively. During pretraining, the model was trained for 2000 epochs using the $\mathcal{L}_{\text{pre}}$. In the fine-tuning phase, 400 epochs were performed using the $\mathcal{L}_{\text{CE}}^{\text{IMG}}$.

4) Metrics: Overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (Kappa) are adopted as the evaluation metrics for quantitative assessment of the classification performance across different methods. To mitigate the impact of randomness, each experiment was independently repeated ten times, and the mean performance along with the standard deviation was reported as the final result.

C. Performance Comparison

Tables IV–VIII present a comparison of classification performance across different methods. On the five target datasets, the proposed PSBL consistently achieves the best performance in terms of OA, AA, and the Kappa. Notably, on the LK, SA, and HC datasets, the OA of the proposed method substantially outperforms that of the comparison approaches.

On the IP dataset, the proposed PSBL achieved state-of-the-art performance, obtaining the highest OA of 79.39%, AA of 87.99%, and Kappa coefficient of 76.74%. While comparison methods, including SSEFN and MEDPL, achieved competitive OA values (78.40% and 71.11%, respectively), their AA remained inferior to PSBL, indicating superior class-balancing capability of our approach. Detailed per-class analysis reveals that PSBL and MEDPL exhibit significant advantages in challenging classes (Classes 4, 12, and 15), where HSIC methods typically underperform due to limited training samples and high spectral similarity among vegetation types. Notably, even in simpler categories where multiple approaches achieve near-perfect accuracy (e.g., Classes 7 and 9), PSBL maintains consistently high and stable performance.

For the LK dataset, PSBL attained optimal comprehensive performance with OA = 64.45%, AA = 75.47%, and Kappa = 59.35%, outperforming the strongest competitor MEDPL by margins of 2.59%, 1.8%, and 2.57%, respectively. Although HSIC methods such as SVM and A^{2}S^{2}K-ResNet exhibited strong performance on specific categories (e.g., Classes 5, 9, and 10), they suffered from significant performance variance across other categories. In contrast, PSBL consistently maintained high accuracy in both dominant and challenging categories (e.g., Classes 1, 3, 4, and 6), demonstrating superior cross-scene generalization capability.

On the SA dataset, PSBL achieved the best comprehensive performance with OA = 95.38%, AA = 96.76%, and Kappa = 94.86%. A class-level analysis further reveals that PSBL attains the highest accuracy in 5 out of the 16 semantic categories, demonstrating particularly strong discriminative capability in challenging classes (e.g., Classes 8 and 15) where competing methods exhibit noticeable performance degradation. While MEDPL and SSEFN demonstrated competitive results in certain categories, they showed inferior consistency. The superiority of PSBL can be attributed to its effective cross-scene semantic knowledge transfer mechanism,

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TABLE IV

CLASSIFICATION RESULTS OF THE IP DATASET (BEST RESULTS ARE SHOWN IN BOLD)

No.	SVM	A^{2}S^{2}K-ResNet	SSTN	SSSAN	DM-MRN	HyMuT	FDFSL	SSEFN	CDMLC	MEDPL	PSBL
1	15.93 ± 5.44	34.21 ± 15.85	63.65 ± 27.44	99.34 ± 1.39	98.05 ± 2.84	38.30 ± 19.19	96.59 ± 2.72	74.18 ± 18.79	98.78±2.11	99.19 ± 1.15	**99.51 ± 0.98**
2	34.24 ± 13.51	72.15 ± 11.01	65.50 ± 14.95	63.26 ± 10.59	43.90 ± 11.80	60.51 ± 9.87	55.62 ± 9.97	**75.00 ± 12.73**	49.83±3.85	73.67 ± 7.42	56.51 ± 10.97
3	32.28 ± 7.66	65.00 ± 23.23	62.63 ± 14.38	47.03 ± 16.63	49.46 ± 7.35	65.50 ± 10.42	64.05 ± 7.26	**70.71 ± 12.00**	64.18±6.24	59.68 ± 11.41	70.58 ± 11.48
4	18.66 ± 4.83	40.97 ± 15.96	48.43 ± 16.26	43.79 ± 23.21	88.64 ± 8.30	78.19 ± 15.15	87.07 ± 7.78	51.97 ± 17.55	84.36±14.66	**99.28 ± 1.02**	94.57 ± 5.40
5	46.02 ± 9.41	87.44 ± 11.16	80.27 ± 21.46	6.36 ± 17.49	64.94 ± 18.31	73.93 ± 7.51	75.17 ± 12.50	**91.89 ± 8.59**	84.36±4.48	73.43 ± 6.44	84.79 ± 4.71
6	77.79 ± 5.55	82.16 ± 14.77	86.42 ± 13.31	85.38 ± 11.41	93.78 ± 4.93	87.45 ± 4.87	83.52 ± 10.22	96.46 ± 3.29	92.04±4.39	**97.15 ± 0.36**	94.18 ± 2.93
7	24.65 ± 11.77	28.45 ± 22.69	24.75 ± 13.33	22.48 ± 15.93	**100.00 ± 0.00**	**100.00 ± 0.00**	99.57 ± 1.30	34.62 ± 17.34	**100.00±0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
8	92.13 ± 4.33	93.11 ± 8.77	98.24 ± 1.75	9.57 ± 5.34	**99.35 ± 0.48**	81.61 ± 15.57	83.11 ± 14.36	98.40 ± 2.58	97.99±1.06	89.78 ± 13.27	97.40 ± 3.78
9	13.89 ± 6.15	15.58 ± 16.15	11.71 ± 11.77	12.68 ± 8.94	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	34.14 ± 13.88	**100.00±0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
10	40.48 ± 6.76	63.15 ± 12.19	65.02 ± 22.24	43.18 ± 14.11	65.83 ± 18.88	68.11 ± 5.28	66.10 ± 13.21	71.85 ± 12.31	60.81±8.13	73.52 ± 0.89	**74.83 ± 3.96**
11	56.42 ± 9.91	77.83 ± 8.06	81.49 ± 9.54	70.58 ± 11.53	61.44 ± 11.39	64.83 ± 8.31	64.51 ± 6.05	**83.79 ± 6.60**	71.15±8.47	63.67 ± 3.19	75.44 ± 6.68
12	23.14 ± 2.74	58.47 ± 12.19	45.34 ± 11.68	38.61 ± 12.91	45.49 ± 7.30	58.78 ± 18.39	59.29 ± 8.11	57.48 ± 15.51	48.47±16.17	71.98 ± 8.49	**72.04 ± 11.02**
13	86.24 ± 10.35	55.95 ± 22.63	83.21 ± 11.28	82.48 ± 8.85	98.73 ± 21.28	98.90 ± 1.11	98.70 ± 1.35	94.45 ± 3.37	99.88±0.22	**100.00 ± 0.00**	99.15 ± 1.36
14	65.13 ± 5.48	92.13 ± 6.53	96.46 ± 2.82	89.82 ± 6.15	92.28 ± 5.44	84.54 ± 16.45	88.60 ± 5.47	96.60 ± 4.45	90.69±5.03	94.97 ± 0.45	**95.20 ± 6.64**
15	36.33 ± 13.75	61.27 ± 13.06	69.17 ± 12.87	51.86 ± 13.18	75.36 ± 8.99	74.02 ± 16.86	80.21 ± 11.64	81.92 ± 9.67	77.10±6.63	75.24 ± 14.37	**94.04 ± 7.62**
16	89.37 ± 12.24	43.79 ± 17.78	50.53 ± 24.82	8.58 ± 11.48	98.27 ± 2.16	98.86 ± 1.02	98.18 ± 1.98	74.19 ± 12.60	**99.72±0.49**	99.24 ± 0.54	99.66 ± 0.73
OA(%)	45.61 ± 3.68	65.12 ± 2.75	66.70 ± 5.88	57.02 ± 3.90	68.94 ± 3.21	70.57 ± 3.47	71.11 ± 2.19	78.40 ± 2.68	72.52±1.07	79.25 ± 0.94	**79.39 ± 2.80**
AA(%)	47.04 ± 2.06	60.73 ± 2.76	64.55 ± 3.71	58.34 ± 4.00	79.98 ± 21.26	80.62 ± 1.91	81.27 ± 1.84	74.23 ± 2.41	82.12±0.62	87.30 ± 0.55	**87.99 ± 1.88**
Kappa	39.49 ± 3.85	61.25 ± 2.79	62.94 ± 6.01	52.07 ± 3.88	65.09 ± 3.56	66.86 ± 3.63	67.55 ± 2.44	75.57 ± 2.94	68.79±0.92	76.04 ± 0.99	**76.74 ± 3.14**

TABLE V

CLASSIFICATION RESULTS OF THE LK DATASET (BEST RESULTS ARE SHOWN IN BOLD)

No.	SVM	A^{2}S^{2}K-ResNet	SSTN	SSSAN	DM-MRN	HyMuT	FDFSL	SSEFN	CDMLC	MEDPL	PSBL
1	30.17 ± 5.87	34.12 ± 19.55	34.61 ± 13.52	31.34 ± 17.68	60.54 ± 6.48	60.99 ± 11.25	63.78 ± 6.08	38.36 ± 11.81	54.42 ± 1.77	70.11 ± 9.48	**71.41 ± 13.45**
2	50.79 ± 23.14	37.55 ± 22.99	63.46 ± 31.04	60.72 ± 25.47	99.85 ± 0.44	98.06 ± 2.77	99.84 ± 0.48	58.49 ± 15.49	86.29 ± 5.65	99.35 ± 1.94	**100.00 ± 0.00**
3	56.57 ± 20.49	44.68 ± 17.90	47.00 ± 11.65	48.60 ± 17.70	52.74 ± 9.50	64.73 ± 13.54	64.06 ± 8.36	69.43 ± 8.12	81.10 ± 4.56	77.17 ± 9.19	**81.23 ± 9.58**
4	8.27 ± 6.02	14.00 ± 5.44	10.64 ± 5.66	13.40 ± 9.10	73.03 ± 14.88	73.38 ± 15.13	68.92 ± 11.75	19.12 ± 6.30	58.11 ± 8.11	70.81 ± 16.23	**92.30 ± 6.11**
5	35.87 ± 7.59	61.57 ± 18.80	58.97 ± 15.05	52.17 ± 22.34	28.58 ± 10.07	46.50 ± 9.08	47.95 ± 11.27	**74.83 ± 10.19**	45.63 ± 0.93	66.96 ± 6.24	66.63 ± 5.90
6	42.01 ± 17.37	16.91 ± 6.00	12.89 ± 4.81	20.63 ± 13.99	41.12 ± 9.49	50.73 ± 14.28	45.46 ± 12.85	16.21 ± 7.40	62.16 ± 2.06	**72.29 ± 11.27**	65.78 ± 10.25
7	41.64 ± 11.47	19.20 ± 5.64	31.56 ± 10.76	30.07 ± 5.89	50.54 ± 15.08	62.14 ± 7.02	58.89 ± 10.50	25.75 ± 7.71	48.99 ± 0.10	**70.34 ± 7.71**	63.01 ± 9.18
8	39.61 ± 4.66	41.64 ± 14.71	40.57 ± 9.53	36.75 ± 11.21	48.68 ± 5.32	**57.41 ± 14.04**	55.65 ± 13.80	43.33 ± 8.55	50.14 ± 17.62	55.49 ± 8.18	54.50 ± 14.53
9	**74.90 ± 7.52**	54.61 ± 6.71	66.54 ± 6.87	59.19 ± 6.40	25.99 ± 8.17	46.38 ± 8.19	38.36 ± 9.20	62.94 ± 4.37	45.26 ± 2.81	40.40 ± 11.49	48.02 ± 6.90
10	**76.53 ± 6.75**	55.22 ± 9.85	59.26 ± 8.40	65.23 ± 10.43	37.60 ± 11.09	55.11 ± 8.23	47.69 ± 8.04	62.35 ± 10.12	52.56 ± 7.95	53.03 ± 9.35	56.15 ± 5.02
11	32.50 ± 5.45	35.25 ± 9.87	31.27 ± 12.36	35.61 ± 10.58	62.42 ± 10.67	69.90 ± 8.52	71.58 ± 6.81	41.96 ± 10.68	66.67 ± 0.00	72.86 ± 10.36	**74.81 ± 11.23**
12	83.31 ± 17.75	71.48 ± 19.08	62.04 ± 19.59	76.27 ± 11.71	76.46 ± 10.74	81.34 ± 10.44	81.62 ± 10.57	69.91 ± 12.62	69.61 ± 0.73	**82.80 ± 11.27**	82.78 ± 9.22
13	**100.00 ± 0.00**	90.29 ± 6.08	99.97 ± 0.07	99.68 ± 0.55	**100.00 ± 0.00**	99.88 ± 0.37	**100.00 ± 0.00**	99.98 ± 0.03	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
14	**100.00 ± 0.00**	82.02 ± 20.13	86.37 ± 6.86	97.84 ± 3.00	99.64 ± 0.59	99.91 ± 0.21	99.75 ± 0.25	96.41 ± 3.16	99.33 ± 0.67	99.80 ± 0.32	**100.00 ± 0.00**
OA(%)	53.64 ± 4.19	45.68 ± 3.22	49.76 ± 4.69	49.27 ± 2.95	46.96 ± 1.84	60.44 ± 2.39	56.51 ± 3.18	56.42 ± 2.00	58.46 ± 0.87	61.76 ± 3.42	**64.45 ± 1.81**
AA(%)	55.16 ± 4.13	47.04 ± 4.18	50.37 ± 3.90	51.96 ± 4.30	60.80 ± 1.71	69.06 ± 1.82	67.40 ± 2.75	55.65 ± 2.03	65.73 ± 2.10	73.67 ± 1.24	**75.47 ± 1.56**
Kappa	48.34 ± 4.18	39.29 ± 3.14	44.06 ± 4.51	43.34 ± 3.14	40.92 ± 1.82	54.77 ± 2.36	50.77 ± 3.14	50.64 ± 2.05	52.47 ± 0.81	56.78 ± 3.27	**59.35 ± 1.84**

TABLE VI

CLASSIFICATION RESULTS OF THE SA DATASET (BEST RESULTS ARE SHOWN IN BOLD)

No.	SVM	A^{2}S^{2}K-ResNet	SSTN	SSSAN	DM-MRN	HyMuT	FDFSL	SSEFN	CDMLC	MEDPL	PSBL
1	94.18 ± 7.93	95.25 ± 6.09	98.39 ± 3.34	98.76 ± 3.01	99.76 ± 0.37	98.87 ± 0.86	98.96 ± 1.63	**100.00 ± 0.00**	99.84±0.14	99.70 ± 0.40	99.55 ± 0.83
2	98.78 ± 0.63	94.94 ± 9.84	98.04 ± 2.41	97.12 ± 2.63	99.90 ± 0.14	98.84 ± 2.22	98.78 ± 3.38	99.83 ± 0.34	**100.00±0.00**	99.76 ± 0.71	99.97 ± 0.06
3	79.57 ± 8.68	94.38 ± 6.87	92.53 ± 8.07	93.84 ± 10.26	94.45 ± 16.08	87.33 ± 11.65	89.10 ± 12.45	**96.94 ± 0.91**	90.98±5.51	96.54 ± 5.98	94.97 ± 5.99
4	96.60 ± 0.77	91.95 ± 3.50	89.35 ± 5.21	95.43 ± 3.48	97.23 ± 0.86	98.67 ± 1.19	99.32 ± 0.55	97.29 ± 1.28	**99.89±0.11**	98.91 ± 2.44	99.56 ± 0.27
5	89.40 ± 8.92	96.90 ± 2.30	95.56 ± 7.29	95.63 ± 6.33	95.93 ± 5.84	91.10 ± 4.25	90.47 ± 6.25	**98.07 ± 1.43**	91.02±2.91	96.82 ± 2.65	94.85 ± 4.03
6	99.87 ± 0.12	97.79 ± 4.96	99.69 ± 0.51	99.32 ± 0.48	97.79 ± 2.64	98.80 ± 1.33	98.41 ± 2.57	99.90 ± 0.16	99.22±1.36	99.67 ± 0.88	**99.97 ± 0.06**
7	94.25 ± 3.02	99.26 ± 0.72	97.45 ± 3.79	99.51 ± 0.32	97.79 ± 2.64	99.10 ± 0.91	98.76 ± 1.37	99.75 ± 0.53	99.82±0.19	99.92 ± 0.10	**99.93 ± 0.08**
8	64.16 ± 7.97	83.13 ± 7.16	79.52 ± 14.28	80.13 ± 9.41	64.19 ± 0.75	75.27 ± 11.44	78.85 ± 3.97	90.25 ± 3.51	81.38±3.00	80.58 ± 3.46	**90.82 ± 5.20**
9	98.57 ± 0.63	97.11 ± 4.83	98.73 ± 0.87	98.42 ± 1.00	99.34 ± 16.29	99.11 ± 1.01	97.98 ± 1.80	99.30 ± 0.24	99.49±0.88	**99.78 ± 0.49**	99.41 ± 1.05
10	80.11 ± 7.02	94.09 ± 3.19	91.54 ± 5.66	94.27 ± 6.49	94.49 ± 1.19	82.28 ± 8.74	85.40 ± 6.40	**96.78 ± 0.79**	89.28±3.76	91.58 ± 3.31	86.54 ± 7.97
11	77.50 ± 9.81	91.59 ± 6.09	90.14 ± 3.15	90.66 ± 9.56	**99.34 ± 0.99**	95.87 ± 4.62	95.51 ± 6.65	93.83 ± 1.56	99.32±0.66	99.14 ± 1.09	97.63 ± 2.86
12	87.90 ± 10.22	96.26 ± 2.84	93.33 ± 7.69	89.38 ± 10.51	97.35 ± 11.44	98.38 ± 2.82	99.64 ± 0.50	98.34 ± 1.48	99.57±0.45	**99.95 ± 0.08**	99.84 ± 0.30
13	78.15 ± 10.64	81.59 ± 16.96	90.89 ± 11.07	92.64 ± 4.42	98.36 ± 1.45	99.26 ± 0.73	98.29 ± 1.23	93.90 ± 4.47	98.14±1.63	97.67 ± 1.65	**99.70 ± 0.44**
14	84.82 ± 12.51	89.84 ± 9.84	90.14 ± 8.94	91.94 ± 5.17	97.89 ± 3.14	97.73 ± 2.15	99.00 ± 0.93	91.37 ± 8.12	**99.48±0.34**	96.91 ± 3.84	98.71 ± 1.13
15	45.21 ± 6.21	66.34 ± 10.07	60.27 ± 13.57	63.18 ± 9.20	82.72 ± 11.39	74.81 ± 5.76	82.53 ± 6.88	64.51 ± 9.04	77.83±8.07	83.44 ± 4.10	**91.84 ± 10.38**
16	47.31 ± 2.04	92.27 ± 8.14	98.64 ± 3.02	98.62 ± 1.02	97.67 ± 2.43	92.28 ± 5.97	92.04 ± 8.10	**99.80 ± 0.48**	94.41±4.33	93.77 ± 4.76	94.83 ± 5.53
OA(%)	78.86 ± 3.16	87.64 ± 2.80	84.04 ± 4.38	87.82 ± 2.66	88.90 ± 2.44	88.63 ± 1.90	90.51 ± 1.17	90.13 ± 0.02	91.32±1.01	92.57 ± 0.57	**95.38 ± 1.71**
AA(%)	85.37 ± 1.52	91.42 ± 1.37	91.51 ± 1.69	92.45 ± 1.54	94.77 ± 1.29	92.98 ± 1.46	93.94 ± 1.26	95.05 ± 0.63	94.98±0.78	96.20 ± 0.68	**96.76 ± 1.26**
Kappa	76.62 ± 3.42	86.31 ± 3.05	82.35 ± 4.75	86.47 ± 2.94	87.71 ± 2.66	87.36 ± 2.08	89.45 ± 1.30	89.35 ± 2.71	90.36±1.12	92.41 ± 0.63	**94.86 ± 1.90**

which successfully adapts spectral bases knowledge learned from source domains to target domains, significantly enhancing spectral discriminability and enabling robust cross-scene generalization.

For the HC dataset, the PSBL demonstrated exceptional performance, achieving top-ranked results with OA = 82.15%, AA = 80.70%, and Kappa = 78.34% among all methods. The framework exhibited particularly strong performance in spectrally complex categories (3, 4, 5, 7, 10, 12, 14, and 15), achieving the best accuracy in 10 out of 16 total classes. Although competitors, including SSEFN, SSTN, and MEDPL, showed competitive results in specific categories (1, 2, 8, 9, 13, and 16), their overall performance remained substantially lower than PSBL.

On the KSC dataset, PSBL achieved state-of-the-art performance with OA = 96.46%, AA = 95.57%, and Kappa = 96.06%. While some competitors excelled in individual categories, PSBL demonstrates superior and more consistent performance across all classes. This balanced proficiency stems from the model's dual capacity to capture long-range spatial dependencies and to integrate transferable spectral-semantic knowledge, leading to robust generalization without category-specific overfitting.

Figs. 7–11 present the classification maps produced by all methods across the five target datasets. Visually, the proposed PSBL yields classification results with noticeably improved region homogeneity and sharper interclass boundaries, while substantially reducing misclassification. This advantage is

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TABLE VII

CLASSIFICATION RESULTS OF THE HC DATASET (BEST RESULTS ARE SHOWN IN BOLD)

No.	SVM	A^2S^2K-ResNet	SSTN	SSSAN	DM-MRN	HyMuT	FDFSL	SSEFN	CDMLC	MEDPL	PSBL
1	83.34 ± 5.66	83.67 ± 8.01	88.37 ± 7.44	73.58 ± 8.26	68.27 ± 8.83	77.72 ± 7.68	71.65 ± 9.40	**92.12 ± 4.44**	77.40 ± 4.98	75.38 ± 10.85	79.52 ± 10.70
2	47.69 ± 14.34	77.84 ± 13.78	74.47 ± 11.39	67.89 ± 15.25	40.54 ± 11.86	45.59 ± 12.16	45.14 ± 7.87	**93.33 ± 2.76**	47.95 ± 8.75	66.37 ± 7.81	58.15 ± 11.14
3	29.71 ± 5.50	46.22 ± 12.04	60.89 ± 17.58	46.83 ± 17.53	44.16 ± 11.46	77.53 ± 5.28	65.41 ± 11.95	70.88 ± 8.75	77.77 ± 3.61	84.17 ± 4.35	**91.90 ± 3.44**
4	29.51 ± 6.97	70.22 ± 12.11	83.46 ± 14.43	57.84 ± 21.84	90.20 ± 4.56	96.16 ± 0.70	88.89 ± 3.71	80.46 ± 10.25	93.55 ± 3.69	97.41 ± 1.46	**97.78 ± 0.57**
5	8.17 ± 1.49	27.04 ± 17.92	17.61 ± 8.58	5.60 ± 3.07	92.29 ± 5.36	85.15 ± 10.83	85.10 ± 13.99	25.88 ± 7.40	86.16 ± 5.65	93.10 ± 6.26	**99.10 ± 1.25**
6	9.92 ± 2.73	31.74 ± 12.54	21.53 ± 5.23	21.47 ± 12.19	40.43 ± 6.63	40.48 ± 10.21	33.51 ± 12.22	38.28 ± 6.75	38.32 ± 5.23	60.14 ± 9.14	**69.59 ± 6.89**
7	20.91 ± 8.33	39.11 ± 15.36	49.90 ± 15.80	26.66 ± 10.27	88.92 ± 5.41	86.20 ± 7.55	76.57 ± 10.36	49.94 ± 8.67	79.54 ± 11.02	84.98 ± 8.54	**94.53 ± 3.22**
8	47.33 ± 19.05	73.96 ± 7.87	68.80 ± 19.61	52.61 ± 17.18	55.29 ± 13.25	41.50 ± 10.48	43.86 ± 9.58	**87.55 ± 6.77**	51.51 ± 15.03	51.93 ± 12.11	76.99 ± 6.79
9	25.37 ± 9.01	59.40 ± 16.99	**72.27 ± 15.54**	41.69 ± 16.42	36.63 ± 8.91	38.79 ± 10.87	26.13 ± 6.38	66.11 ± 8.26	42.31 ± 10.39	57.74 ± 14.07	68.35 ± 7.42
10	45.33 ± 19.28	58.12 ± 19.32	58.86 ± 30.24	69.13 ± 18.92	77.53 ± 16.16	83.61 ± 10.59	66.48 ± 16.81	83.78 ± 14.40	84.73 ± 5.62	85.79 ± 12.48	**88.95 ± 7.34**
11	41.65 ± 18.99	46.67 ± 17.21	57.83 ± 33.53	43.11 ± 9.57	85.83 ± 11.95	66.87 ± 15.79	48.30 ± 18.22	53.37 ± 12.00	72.59 ± 17.23	**89.64 ± 5.65**	83.52 ± 13.75
12	8.28 ± 3.89	26.37 ± 14.31	18.84 ± 6.50	13.50 ± 2.47	60.30 ± 14.25	53.65 ± 9.15	47.43 ± 15.41	45.89 ± 17.02	55.55 ± 9.40	74.58 ± 9.99	**92.10 ± 8.49**
13	18.74 ± 7.10	27.75 ± 13.63	31.52 ± 12.44	19.37 ± 6.51	27.94 ± 8.84	32.48 ± 10.95	29.07 ± 6.40	56.33 ± 14.74	34.35 ± 7.53	55.62 ± 7.71	**57.24 ± 11.72**
14	71.12 ± 8.80	63.34 ± 17.21	69.97 ± 19.69	59.16 ± 15.06	47.88 ± 12.10	62.32 ± 12.88	55.64 ± 9.85	69.45 ± 11.92	54.40 ± 10.30	62.47 ± 12.61	**73.47 ± 5.77**
15	24.86 ± 11.09	43.90 ± 27.79	35.82 ± 10.66	20.46 ± 7.76	79.92 ± 6.71	72.27 ± 7.03	69.51 ± 11.22	44.93 ± 24.06	76.94 ± 11.58	81.29 ± 8.80	**84.01 ± 7.05**
16	92.73 ± 9.55	92.86 ± 5.89	**99.19 ± 1.68**	95.39 ± 4.28	87.93 ± 8.99	90.77 ± 9.19	93.18 ± 8.93	97.11 ± 4.20	86.10 ± 12.68	85.46 ± 12.56	95.33 ± 7.78
OA(%)	47.76 ± 2.56	60.33 ± 5.42	65.11 ± 6.04	53.39 ± 3.65	67.67 ± 3.30	71.14 ± 5.28	66.85 ± 4.26	76.33 ± 3.24	70.49 ± 2.46	75.84 ± 3.62	**82.15 ± 3.36**
AA(%)	37.80 ± 3.27	54.26 ± 3.57	56.83 ± 4.65	44.64 ± 3.87	64.00 ± 1.41	65.69 ± 2.96	59.12 ± 3.50	65.96 ± 3.03	66.20 ± 1.04	75.38 ± 2.02	**80.70 ± 2.34**
Kappa	41.23 ± 2.69	54.73 ± 5.50	60.19 ± 6.58	47.20 ± 3.73	62.92 ± 3.52	66.81 ± 5.68	61.74 ± 4.55	72.64 ± 3.69	66.14 ± 2.46	72.35 ± 3.90	**78.34 ± 2.76**

TABLE VIII

CLASSIFICATION RESULTS OF THE KSC DATASET (BEST RESULTS ARE SHOWN IN BOLD)

No.	SVM	A^2S^2K-ResNet	SSTN	SSSAN	DM-MRN	HyMuT	FDFSL	SSEFN	CDMLC	MEDPL	PSBL
1	93.13 ± 2.02	**98.94 ± 1.55**	98.42 ± 2.62	98.71 ± 1.09	98.04 ± 1.79	95.56 ± 3.32	98.45 ± 1.45	98.74 ± 1.17	97.88 ± 1.53	97.61 ± 1.91	98.57 ± 1.95
2	73.05 ± 10.91	91.28 ± 8.03	88.76 ± 7.92	86.76 ± 6.72	76.38 ± 10.32	85.38 ± 11.64	87.77 ± 10.10	95.28 ± 3.97	81.85 ± 9.39	**96.22 ± 5.37**	93.95 ± 9.73
3	55.98 ± 6.47	90.77 ± 7.59	82.37 ± 13.71	76.01 ± 9.28	92.50 ± 5.69	88.29 ± 11.62	87.09 ± 8.31	87.72 ± 6.57	85.42 ± 5.03	92.99 ± 11.24	**96.61 ± 4.81**
4	34.85 ± 8.62	80.62 ± 9.31	47.62 ± 30.54	72.90 ± 9.30	47.46 ± 10.45	**96.16 ± 0.70**	64.13 ± 15.22	88.51 ± 6.85	58.14 ± 7.19	82.02 ± 12.66	80.24 ± 18.16
5	36.14 ± 16.14	90.49 ± 11.96	62.21 ± 25.49	67.23 ± 11.61	63.85 ± 11.06	78.33 ± 5.91	80.83 ± 13.38	78.23 ± 8.90	67.82 ± 15.94	90.13 ± 7.71	**94.10 ± 9.34**
6	45.35 ± 12.26	83.43 ± 17.68	73.58 ± 23.98	80.82 ± 11.69	85.55 ± 10.72	92.68 ± 7.23	92.95 ± 4.46	93.46 ± 7.27	95.09 ± 3.62	94.51 ± 10.44	**95.94 ± 11.46**
7	64.85 ± 10.33	80.70 ± 11.86	63.81 ± 28.40	73.66 ± 10.33	99.14 ± 1.56	91.80 ± 9.66	96.80 ± 7.07	80.45 ± 14.33	93.40 ± 4.96	98.40 ± 3.32	**100.00 ± 0.00**
8	55.67 ± 8.60	94.17 ± 5.62	82.47 ± 15.11	90.09 ± 9.15	85.59 ± 12.39	90.33 ± 6.18	95.00 ± 3.93	**96.18 ± 5.06**	82.30 ± 8.57	90.26 ± 12.96	95.26 ± 4.93
9	71.27 ± 6.77	95.91 ± 6.61	93.74 ± 9.12	94.48 ± 7.39	91.60 ± 6.97	88.78 ± 13.77	92.52 ± 6.49	**96.05 ± 7.46**	95.38 ± 7.35	91.73 ± 8.29	91.59 ± 8.19
10	81.08 ± 14.23	97.95 ± 2.86	99.87 ± 0.32	99.43 ± 0.98	81.91 ± 12.82	98.70 ± 0.86	98.57 ± 1.94	**99.92 ± 0.23**	98.45 ± 1.06	99.87 ± 0.26	99.02 ± 2.10
11	88.96 ± 4.80	96.82 ± 5.65	97.37 ± 2.37	98.00 ± 2.17	98.85 ± 1.92	95.12 ± 4.35	97.90 ± 3.15	98.97 ± 0.47	97.49 ± 2.15	98.02 ± 2.55	**99.88 ± 0.19**
12	79.83 ± 13.38	97.33 ± 2.30	85.44 ± 28.80	94.47 ± 5.18	92.78 ± 5.80	95.46 ± 2.59	94.30 ± 3.84	**98.95 ± 1.55**	91.45 ± 4.67	95.38 ± 2.13	97.27 ± 2.21
13	99.51 ± 1.29	99.84 ± 0.32	99.31 ± 0.95	98.47 ± 1.39	99.76 ± 0.68	**100.00 ± 0.00**	**100.00 ± 0.00**	99.98 ± 0.07	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
OA(%)	74.59 ± 2.43	94.21 ± 2.50	87.16 ± 8.24	91.39 ± 1.06	89.67 ± 1.70	92.40 ± 2.12	93.91 ± 0.72	95.52 ± 1.23	91.73 ± 0.76	95.45 ± 1.20	**96.46 ± 1.30**
AA(%)	67.67 ± 2.70	92.17 ± 2.36	82.69 ± 6.41	87.00 ± 1.53	85.65 ± 1.94	89.60 ± 1.66	91.26 ± 1.07	93.26 ± 1.69	88.05 ± 0.76	94.40 ± 1.57	**95.57 ± 1.63**
Kappa	71.76 ± 2.67	93.56 ± 2.77	85.76 ± 9.01	90.41 ± 1.19	88.50 ± 1.89	91.54 ± 2.34	93.22 ± 0.80	95.21 ± 1.50	90.79 ± 0.84	94.93 ± 1.34	**96.06 ± 1.45**

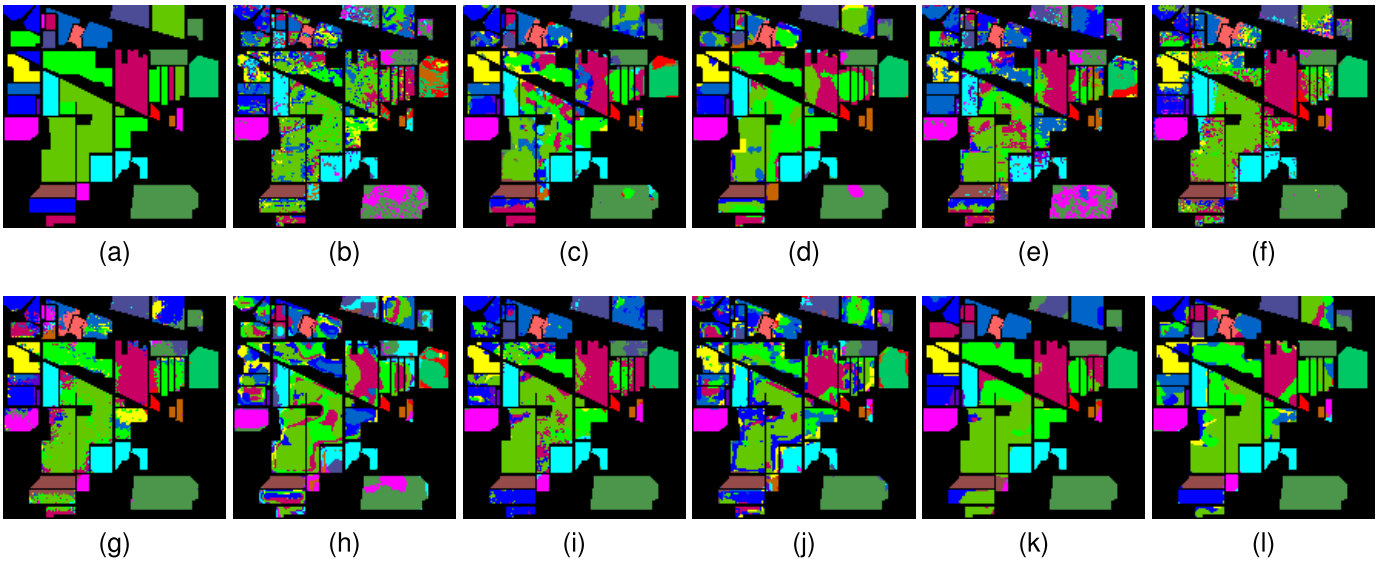


Fig. 7. Classification maps for the IP dataset. (a) GT. (b) SVM. (c) A^2S^2K-ResNet. (d) SSTN. (e) SSSAN. (f) DM-MRN. (g) HyMuT. (h) FDFSL. (i) SSEFN. (j) CDMLC. (k) MEDPL. (l) PSBL.

particularly evident in the agriculturally structured areas of the IP and SA datasets, where PSBL maintains stronger spatial consistency compared to other methods that exhibit pronounced misclassification. These visual improvements align closely with the quantitative accuracy metrics reported earlier, confirming that the integration of cross-scene spectral knowledge enhances discriminability between spectrally similar vegetation types. Notably, PSBL demonstrates superior robustness against overfitting compared to methods like MEDPL and SSEFN that produce fragmented classification maps. This stems from its ability to mitigate overfitting by capturing long-range dependencies.

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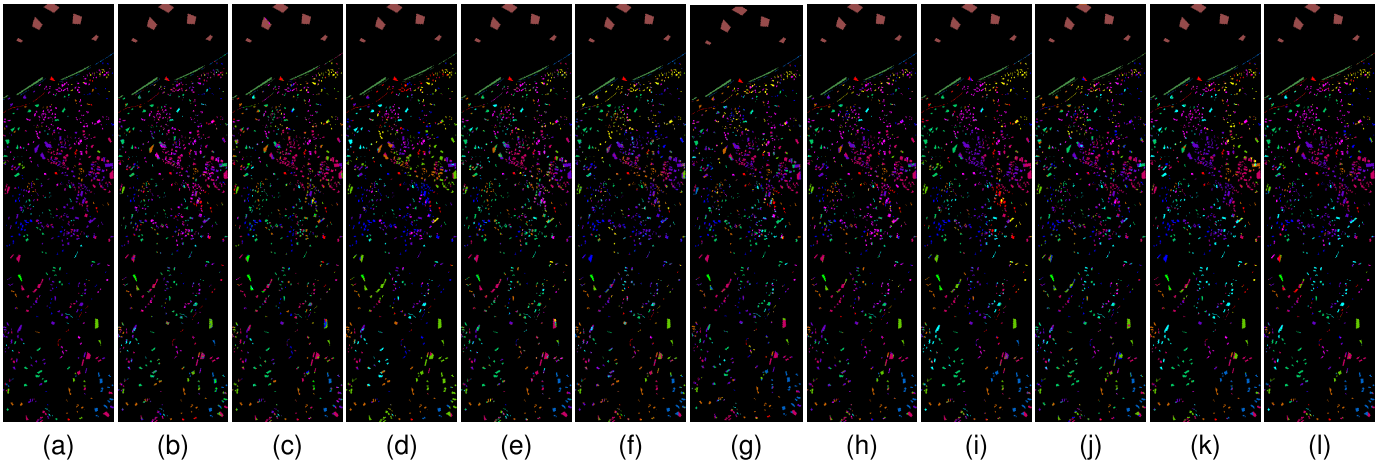


Fig. 8. Classification maps for the LK dataset. (a) GT. (b) SVM. (c) A^2S^2K-ResNet. (d) SSTN. (e) SSSAN. (f) DM-MRN. (g) HyMuT. (h) FDFSL. (i) SSEFN. (j) CDMLC. (k) MEDPL. (l) PSBL.

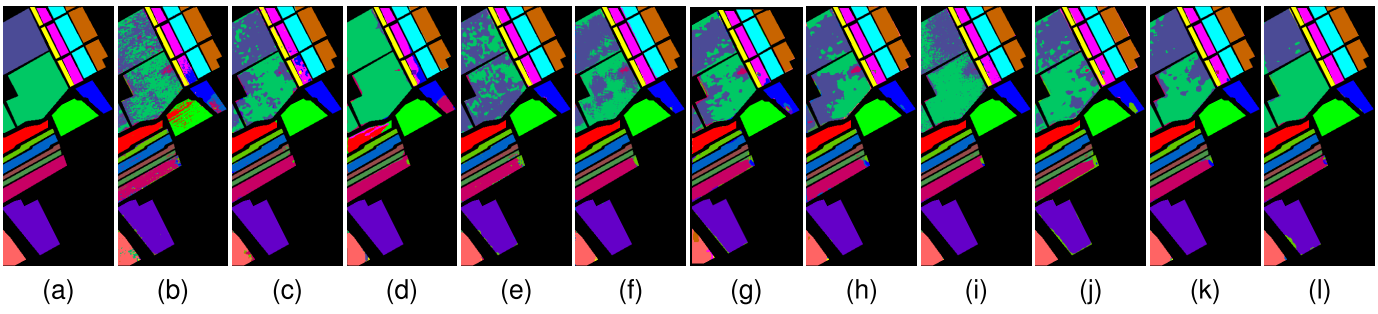


Fig. 9. Classification maps for the SA dataset. (a) GT. (b) SVM. (c) A^2S^2K-ResNet. (d) SSTN. (e) SSSAN. (f) DM-MRN. (g) HyMuT. (h) FDFSL. (i) SSEFN. (j) CDMLC. (k) MEDPL. (l) PSBL.

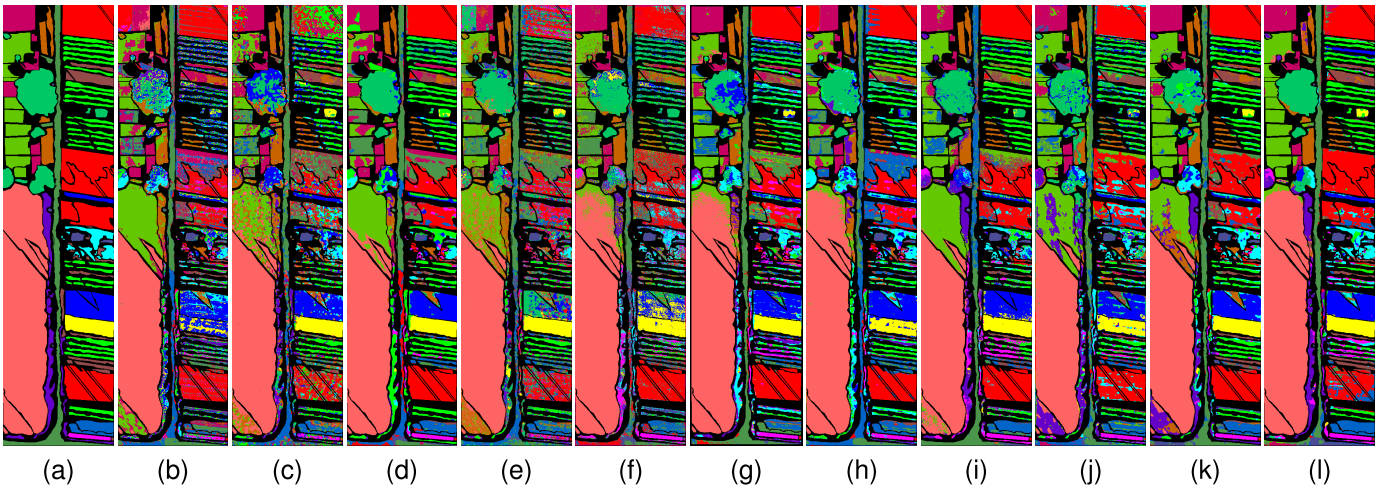


Fig. 10. Classification maps for the HC dataset. (a) GT. (b) SVM. (c) A^2S^2K-ResNet. (d) SSTN. (e) SSSAN. (f) DM-MRN. (g) HyMuT. (h) FDFSL. (i) SSEFN. (j) CDMLC. (k) MEDPL. (l) PSBL.

D. Impact of the Number of Training Samples

To evaluate the robustness of the proposed method, we compared its performance against SSTN, A^2S^2K-ResNet, SSSAN, DM-MRN, FDFSL, and SSEFN on the IP, SA, and KSC datasets. Experiments were conducted under varying numbers of labeled samples per class $L \in \{1, 2, 3, 4, 5\}$, using OA as the evaluation metric. As illustrated in Fig. 12, classification accuracy improves across all methods as the number of training samples increases. Notably, once the number of training samples exceeds two, PSBL achieves a consistent and clear performance margin over all competing methods on all three datasets. In contrast, other methods exhibit slower performance growth and higher sensitivity to small sample sizes. Although SSEFN and FDFSL show competitive results at larger L values, they still fall short of the

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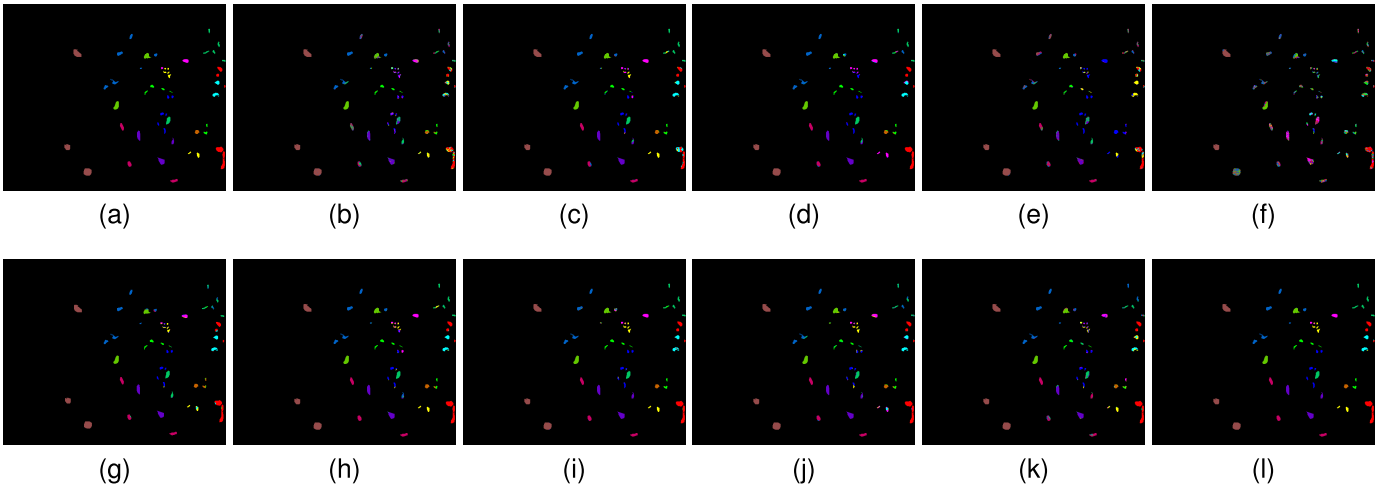


Fig. 11. Classification maps for the KSC dataset. (a) GT. (b) SVM. (c) A^{2}S^{2}K-ResNet. (d) SSTN. (e) SSSAN. (f) DM-MRN. (g) HyMuT. (h) FDFSL. (i) SSEFN. (j) CDMLC. (k) MEDPL. (l) PSBL.

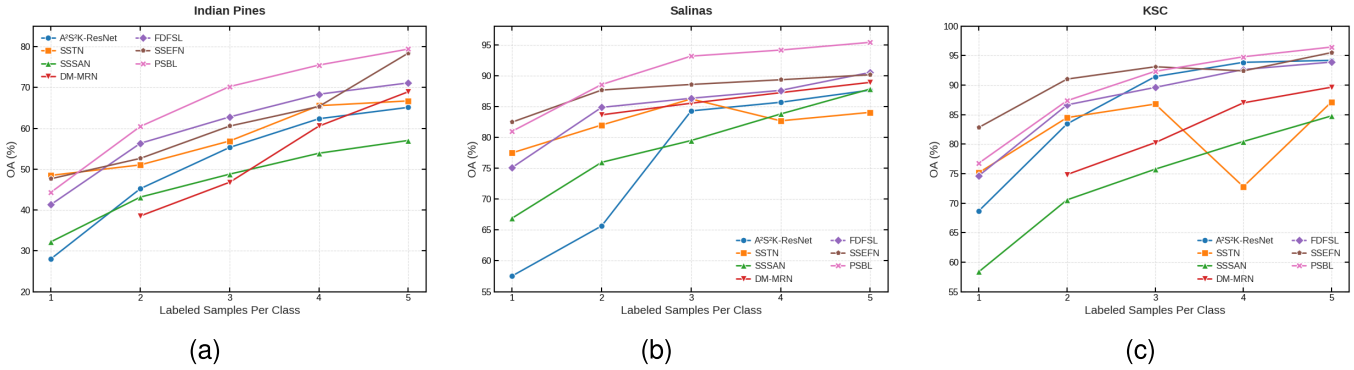


Fig. 12. Performance comparison across different datasets with varying numbers of training samples. (a) IP dataset. (b) SA dataset. (c) KSC dataset.

TABLE IX
PERFORMANCE COMPARISON WITH DIFFERENT COMPONENTS

L	Baseline	Transformer	Vision-Mamba	Cross-Mamba	Calibration	Datasets				
						IP	SA	LK	HC	KSC
5	✓					74.90 ± 3.07	88.05 ± 2.42	58.62 ± 2.26	75.45 ± 3.54	87.85 ± 3.87
	✓	✓			✓	76.49 ± 3.09	93.75 ± 1.71	64.15 ± 2.18	80.10 ± 2.63	95.37 ± 1.23
	✓		✓		✓	77.38 ± 4.25	93.64 ± 1.52	63.45 ± 3.33	81.13 ± 2.27	95.84 ± 1.93
	✓			✓		78.19 ± 2.18	93.60 ± 2.42	62.14 ± 3.02	81.22 ± 3.11	95.05 ± 2.12
	✓				✓	74.84 ± 3.03	88.64 ± 1.93	64.36 ± 1.95	77.56 ± 1.92	92.72 ± 1.92
	✓			✓	✓	$\mathbf{79.39 \pm 2.80}$	$\mathbf{95.38 \pm 1.17}$	$\mathbf{64.45 \pm 1.81}$	$\mathbf{82.15 \pm 3.36}$	$\mathbf{96.46 \pm 1.30}$

accuracy and stability consistently delivered by PSBL. Collectively, these results confirm that PSBL performs strongly not only under the five-shot setting but also remains robust in more challenging low-shot scenarios, underscoring its generalization capacity in CSFS-HSIC.

E. Ablation Study

To evaluate the individual and joint contributions of different components within the PSBL, we conducted a series of ablation studies. Specifically, we systematically removed the cross-Mamba and the calibration module from the PSBL backbone. In addition, we replaced cross-Mamba with the Transformer in the KCCM to demonstrate the superiority of our design.

1) Effect of Calibration: The calibration module operates by infusing feature representations with transferable semantic knowledge across diverse scenes. As evidenced in Table IX, integrating the calibration module with the baseline model yields substantial performance gains, particularly on datasets characterized by significant spectral variability. For instance, on the LK dataset, the OA improves from 58.62% to 64.36%. These enhancements demonstrate that incorporating cross-scene semantic knowledge effectively strengthens generalization capabilities, especially under conditions of substantial spectral distribution shift.

2) Effect of Cross-Mamba: The cross-Mamba module is designed to efficiently capture long-range spatial dependencies. Comparative experiments reveal that this module delivers

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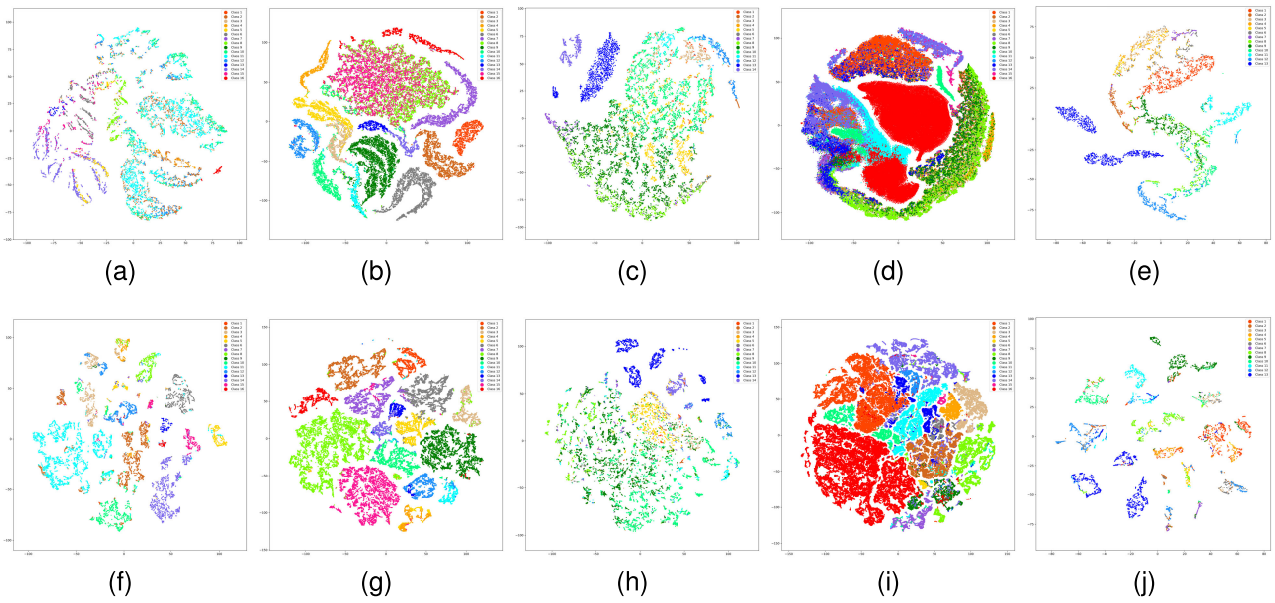


Fig. 13. Feature visualization using t-SNE on target datasets. Visualization of raw features on (a) IP dataset, (b) SA dataset, (c) LK dataset, (d) HC dataset, and (e) KSC dataset. Visualization of embedded features learned by PSBL on (f) IP dataset, (g) SA dataset, (h) LK dataset, (i) HC dataset, and (j) KSC dataset.

TABLE X
IMPACT OF PRETRAINING ON THE PERFORMANCE OF BASELINE METHODS AND THE PROPOSED PSBL

L	Method	Pretraining	Datasets				
			IP	SA	LK	HC	KSC
5	Baseline		67.10 ± 2.48	86.34 ± 0.78	58.22 ± 2.18	68.88 ± 2.82	85.01 ± 3.31
		✓	$\mathbf{74.90 \pm 3.07}$	$\mathbf{88.05 \pm 2.42}$	$\mathbf{58.62 \pm 2.26}$	$\mathbf{75.45 \pm 3.54}$	$\mathbf{87.85 \pm 3.87}$
5	PSBL		77.67 ± 2.49	93.83 ± 1.32	58.98 ± 3.10	79.97 ± 2.73	94.11 ± 2.90
		✓	$\mathbf{79.39 \pm 2.80}$	$\mathbf{95.38 \pm 1.17}$	$\mathbf{64.45 \pm 1.81}$	$\mathbf{82.15 \pm 3.36}$	$\mathbf{96.46 \pm 1.30}$

consistent improvements in OA across all benchmark datasets when compared to the baseline model. Taking the HC dataset as an example, the OA increases from 75.45% to 81.22%. More notably, in direct comparisons with the Transformer architecture, cross-Mamba achieves comparable or superior performance: it attains an OA of 78.19% on the IP dataset, outperforming Transformer's 76.49%, and achieves 81.22% on the HC dataset, exceeding Transformer's 80.10%. These results validate its enhanced capability in modeling long-range spatial dependencies.

3) Effect of KCCM: Furthermore, the integration of the calibration and cross-Mamba modules yields the KCCM module. As demonstrated in Table IX, the concurrent utilization of both components achieves the optimal OA, confirming the effectiveness of KCCM and demonstrating the complementary effects between the two components. In addition, the t-SNE visualization in Fig. 13 compares the feature embeddings learned by PSBL with the raw feature, unequivocally demonstrating that PSBL substantially enhances feature separability, thereby validating the efficacy of the proposed framework.

4) Effect of Pretraining: The experimental results presented in Table X provide compelling evidence for the effectiveness of the pretraining strategy across both the baseline method and our proposed PSBL. Both the baseline and PSBL show clear performance gains across all five datasets when pretraining is applied. Our method achieves optimal results with pretraining, notably on challenging datasets like LK (+5.47%) and KSC (+2.35%). The combination of our architecture and pretraining exhibits a synergistic effect, yielding superior OA and improved stability, as evidenced by reduced standard deviations. These results validate that pretraining is a crucial strategy for enhancing generalization in HSIC.

5) Effect of Discriminative Loss: The experimental results presented in Table XI provide compelling evidence for the effectiveness of our proposed loss configurations during the pretraining stage. Compared to using only the $\mathcal{L}_{\mathrm{CE}}^{\mathrm{IMG}}$, the addition of the $\mathcal{L}_{\mathrm{CE}}^{\mathrm{PB}}$ contributes notably to performance enhancement, particularly on complex datasets. The OA on the IP dataset improves from 77.62% to 78.29%, while the HC dataset achieves a more substantial gain from 80.43% to 81.84%. These improvements validate the effectiveness of this loss function in enhancing interclass discriminability and feature representation quality.

6) Effect of Reconstructive Loss: Compared to using only the $\mathcal{L}_{\mathrm{CE}}^{\mathrm{IMG}}$, the introduction of $\mathcal{L}_{\mathrm{REC}}^{\mathrm{PB}}$ achieves the highest individual improvement on the IP dataset (78.74%), representing a 1.12% point gain over the baseline. This performance advantage underscores the importance of preserving spectral integrity through reconstruction constraints during pretraining. The combined use of all three losses achieves optimal

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TABLE XI
PERFORMANCE COMPARISON WITH DIFFERENT LOSS CONFIGURATIONS

L	$\mathcal{L}_{CE}^{IMG}$	$\mathcal{L}_{CE}^{PB}$	$\mathcal{L}_{REC}^{PB}$	Datasets				
				IP	SA	LK	HC	KSC
5	✓			77.62 ± 3.53	95.15 ± 1.67	$\mathbf{64.49 \pm 2.96}$	80.43 ± 2.89	$\mathbf{96.56 \pm 1.05}$
	✓	✓		78.29 ± 2.72	94.71 ± 1.79	64.22 ± 2.82	81.84 ± 3.79	96.16 ± 1.32
	✓		✓	78.74 ± 3.52	94.96 ± 1.72	63.42 ± 2.51	81.08 ± 2.95	95.80 ± 1.81
	✓	✓	✓	$\mathbf{79.39 \pm 2.80}$	$\mathbf{95.38 \pm 1.17}$	64.45 ± 1.81	$\mathbf{82.15 \pm 3.36}$	96.46 ± 1.30

TABLE XII
PERFORMANCE COMPARISON WITH DIFFERENT PATCH SIZES (S) OF PROTOTYPICAL SPECTRA

S	Datasets				
	IP	SA	LK	HC	KSC
1	79.39 ± 2.80	$\mathbf{95.38 \pm 1.17}$	$\mathbf{64.45 \pm 1.81}$	$\mathbf{82.15 \pm 3.36}$	96.46 ± 1.30
3	78.03 ± 2.96	94.01 ± 1.86	63.31 ± 2.20	81.01 ± 2.92	$\mathbf{96.59 \pm 1.47}$
5	79.13 ± 2.57	94.20 ± 1.23	63.73 ± 2.85	81.14 ± 3.65	96.30 ± 1.11
7	$\mathbf{79.47 \pm 2.98}$	94.44 ± 1.40	63.93 ± 2.31	81.18 ± 3.81	96.45 ± 1.28
9	78.75 ± 3.75	94.09 ± 2.22	64.30 ± 2.29	80.84 ± 2.62	96.51 ± 1.30

performance on multiple datasets (IP: 79.39%, SA: 95.38%, and HC: 82.15%) with improved stability, as seen in the reduced standard deviation on LK (± 1.81 versus ± 2.96 baseline). These results affirm the efficacy of a balanced multiloss pretraining strategy for robust CSFS-HSIC.

F. Parameter Sensitivity Analysis

In this section, we investigate the impact of key hyperparameters on model performance under the five-shot setting. We focus on two parameters: the patch size (S) for prototypical spectra inputs to the PSBLN, and the number of compressed prototypical spectra (M). Notably, no dataset-specific parameter tuning was performed during testing; all experiments used a unified parameter configuration.

1) Patch Size (S): The S was evaluated to examine the effect of neighborhood size on the generation of spectral bases, with the corresponding results reported in Table XII. We observed that the best classification performance was achieved with $S = 1$ for the SA, LK, and HC datasets. This result implies that the PSBLN relies predominantly on spectral information from the central pixel, with minimal contribution from surrounding spatial context, during the spectral bases generation process. Consequently, the input to the PSBLN was defined using class-wise mean spectra, thereby focusing the model on distilling transferable spectral knowledge across scenes.

2) Number of Compressed Prototypical Spectra (M): M denotes the number of compressed prototypical spectra and also the number of basis vectors for each spectral basis. To analyze the sensitivity of PSBL to M, we evaluated the OA of the proposed PSBL for different values of M on the IP, LK, SA, HC, and KSC datasets. Specifically, we systematically investigated the impact of M on performance by testing a series of M values ($M \in \{0, 10, 30, 70, 100, 140, 200\}$). Here, $M = 140$ corresponds to direct spectral mapping, $M > 140$ indicates category expansion, and $M < 140$ indicates category

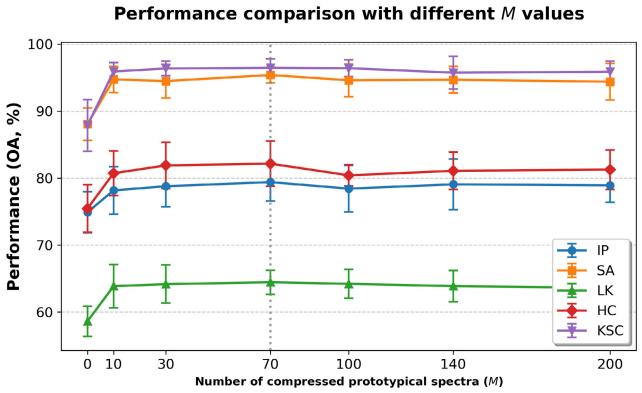


Fig. 14. Performance varies with the number of compressed prototypical spectra in PSBLN.

TABLE XIII
PERFORMANCE COMPARISON WITH DIFFERENT NUMBERS OF COMPRESSED PROTOTYPICAL SPECTRA (M)

M	Datasets				
	IP	SA	LK	HC	KSC
0	74.90 ± 3.07	88.05 ± 2.42	58.62 ± 2.26	75.45 ± 3.54	87.85 ± 3.87
10	78.16 ± 3.55	94.72 ± 1.95	63.86 ± 3.24	80.72 ± 3.33	95.90 ± 1.36
30	78.78 ± 3.05	94.47 ± 2.53	64.16 ± 2.85	81.89 ± 3.47	96.36 ± 1.07
70	$\mathbf{79.39 \pm 2.80}$	$\mathbf{95.38 \pm 1.17}$	$\mathbf{64.45 \pm 1.81}$	$\mathbf{82.15 \pm 3.36}$	$\mathbf{96.46 \pm 1.30}$
100	78.42 ± 3.50	94.59 ± 2.43	64.21 ± 2.14	80.41 ± 1.58	96.40 ± 1.23
140	79.06 ± 3.79	94.66 ± 1.99	63.87 ± 2.33	81.08 ± 2.80	95.73 ± 2.44
200	78.91 ± 2.53	94.39 ± 2.74	63.61 ± 1.56	81.26 ± 2.94	95.87 ± 1.59

compression. When $M = 0$, this is equivalent to removing the PSBLN and all KCCM modules. As shown in Fig. 14, PSBL performs worst at $M = 0$ across all datasets, which validates the necessity of spectral basis learning. For the IP, LK, HC, and KSC datasets, when $M \leq 70$, the overall OA improves as M increases, reaching its optimal value at $M = 70$, indicating that moderate class compression can effectively mitigate spectral redundancy. When $70 < M \leq 200$, PSBL's performance declines across all datasets, which may be attributed to excessively large M values introducing too many parameters, thereby degrading generalization performance. For all datasets, the specific performance metrics corresponding to different M values can be found in Table XIII. Overall, the proposed method demonstrates excellent performance under moderate class compression.

3) Number of KCCM Modules: The KCCM module is capable of seamlessly injecting transferable semantic knowledge and long-range spatial dependencies into feature representations. To determine the optimal number of KCCM

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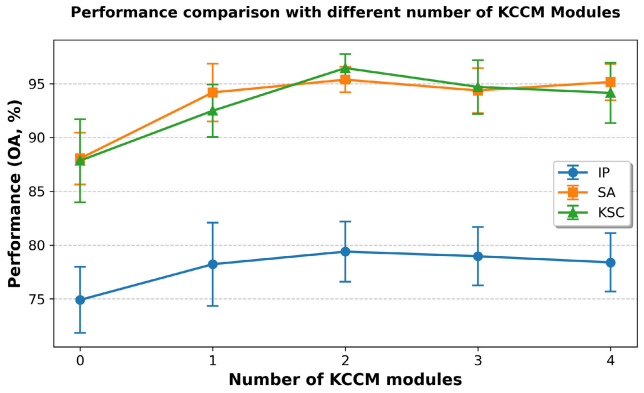


Fig. 15. Performance varies with the number of KCCM modules of the proposed PSBL.

TABLE XIV
COMPLEXITY AND TESTING TIME OF THE PROPOSED PSBL AND OTHER METHODS

Datasets	Metrics	FDFSL	SSEFN	PSBL
IP	Params (M)	**0.19**	16.35	0.23
	GFLOPs	536.13	4013.45	**3.83**
	GPU memory (M)	5803.28	1886.20	**833.37**
	Training time (s)	1167.07	1173.67	**107.249**
	Testing time (s)	10.73	151.07	**0.05**
SA	Params (M)	**0.19**	16.49	0.23
	GFLOPs	2851.36	3392.02	**18.82**
	GPU memory (M)	5806.37	**1891.33**	3713.23
	Training time (s)	1225.49	1126.63	**195.50**
	Testing time (s)	33.11	837.17	**0.14**
KSC	Params (M)	**0.18**	15.47	0.23
	GFLOPs	270.31	263.69	**51.49**
	GPU memory (M)	5745.99	**1856.51**	10003.53
	Training time (s)	1628.98	1766.74	**387.15**
	Testing time (s)	4.94	75.89	**0.18**

modules in the proposed PSBL, we investigate the variation in OA by integrating 0–4 KCCM modules with a step size of 1 on the IP, SA, and KSC datasets. The results are illustrated in Fig. 15. It can be observed that the performance is worse when no KCCM module is integrated across all three datasets. As the number of KCCM modules increases, the performance of PSBL improves accordingly. The optimal performance is achieved when two KCCM modules are integrated, after which the performance begins to decline. Therefore, PSBL ultimately integrates two KCCM modules in total.

G. Computational Complexity

In this section, we analyze the computational complexity of the proposed methods. It is worth noting that, as a patch-free method [45], PSBL was evaluated against the two patch-based methods with higher computational complexity—FDFSL and SSEFN—for the IP, SA, and KSC datasets. The comparison metrics included the number of parameters, gigaflops (GFLOPs), GPU memory usage during inference, training time, and testing time. Note that for PSBL, since the model requires only a single pretraining session before it can be transferred to any target image for fine-tuning, we did not include the model pretraining time in the reported training time (pretraining time on a single NVIDIA GeForce RTX 3090 GPU is approximately 60 h). To ensure fairness, the batch size for patch-based methods was uniformly set to 1024 to facilitate parallel computation. As shown in Table XIV, PSBL demonstrates significantly lower parameter counts and computational costs across all datasets. For example, on the SA dataset, PSBL required only 18.82 GFLOPs, representing a reduction in computational effort of more than two orders of magnitude compared to FDFSL (2851.36 GFLOPs) and SSEFN (3392.02 GFLOPs). Regarding training time, the proposed PSBL achieves the shortest training time across all datasets. In terms of inference efficiency, PSBL also demonstrates significant advantages: on the IP dataset, it completes the test in just 0.05 s, whereas FDFSL and SSEFN require 10.73 and 151.07 s, respectively. This trend holds true on the SA and KSC datasets as well, where PSBL reduces inference latency by one to three orders of magnitude. Regarding GPU memory, the proposed PSBL consumes 833.37 MB of GPU memory on the IP dataset, the lowest among all methods. On the SA and KSC datasets, SSEFN has the lowest GPU memory consumption, while PSBL's memory consumption is higher than that of the other two methods. These results highlight PSBL's exceptional computational efficiency. By eliminating patch sampling and implementing efficient modeling of long-range spatial dependencies, PSBL achieves lightweight computational complexity and accelerated inference capabilities, making it particularly suitable for real-time, large-scale applications such as FS-HSIC in UAV or satellite remote sensing scenarios.

V. CONCLUSION

This article first formalizes the CSFS-HSIC task and proposes a novel PSBL approach to address it. The method comprises two subnetworks: PSBLN and KECHN. PSBLN extracts cross-scene transferable semantic knowledge from the spectral domain of auxiliary hyperspectral datasets. KECHN performs knowledge-enhanced HSI classification through multiple KCCM modules, each integrating a calibration submodule and a cross-Mamba submodule. This seamlessly injects cross-scene transferable spatial semantic knowledge and long-range spatial dependencies into feature representations, enhancing generalization capabilities. Furthermore, we establish a standardized CSFS-HSIC evaluation benchmark to validate the effectiveness of the proposed method. Extensive experiments across five benchmark datasets consistently demonstrate state-of-the-art performance, confirming the effectiveness of the proposed approach. While PSBL delivers satisfactory results, it shares limitations common to most existing methods when confronted with extreme spectral variation: additional data augmentation strategies are required to improve robustness. The method's performance also partly depends on the scale and quality of the source-domain auxiliary data. Future research could explore semi-supervised or unsupervised learning paradigms to reduce reliance on large-scale annotated source data. Moreover, as a patch-free FS-HSIC method, PSBL currently imposes relatively high hardware requirements; investigating model compression techniques could enhance deployment efficiency in practical scenarios. We believe that addressing these limitations in future work will further improve the robustness and applicability of the proposed framework, making it more suitable for real-world HSI analysis tasks.

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REFERENCES

[1] J. T. Daly et al., "Tunable narrow-band filter for LWIR hyperspectral imaging," in *Proc. SPIE*, vol. 3948, pp. 104–115, Apr. 2000.

[2] L. Tu, X. Huang, J. Li, J. Yang, and J. Gong, "A multi-task learning method for extraction of newly constructed areas based on bi-temporal hyperspectral images," *ISPRS J. Photogramm. Remote Sens.*, vol. 208, pp. 308–323, Feb. 2024.

[3] Y. Huang et al., "Cross-scene wetland mapping on hyperspectral remote sensing images using adversarial domain adaptation network," *ISPRS J. Photogramm. Remote Sens.*, vol. 203, pp. 37–54, Sep. 2023.

[4] D. Tuia, R. Flamary, and N. Courty, "Multiclass feature learning for hyperspectral image classification: Sparse and hierarchical solutions," *ISPRS J. Photogramm. Remote Sens.*, vol. 105, pp. 272–285, Jul. 2015.

[5] F. Ullah, I. Ullah, R. U. Khan, S. Khan, K. Khan, and G. Pau, "Conventional to deep ensemble methods for hyperspectral image classification: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3878–3916, 2024.

[6] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[7] M. Liang et al., "An efficient transformer with neighborhood contrastive tokenization for hyperspectral images classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 131, Jul. 2024, Art. no. 103979.

[8] C. Wang, J. Huang, M. Lv, H. Du, Y. Wu, and R. Qin, "A local enhanced mamba network for hyperspectral image classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 133, Sep. 2024, Art. no. 104092.

[9] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[10] S. Liu, C. Fu, Y. Duan, X. Wang, and F. Luo, "Spatial–spectral enhancement and fusion network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5502414.

[11] J. Zeng, Z. Xue, L. Zhang, Q. Lan, and M. Zhang, "Multistage relation network with dual-metric for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510017.

[12] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[13] X. Zuo, X. Yu, B. Liu, P. Zhang, and X. Tan, "FSL-EGNN: Edge-labeling graph neural network for hyperspectral image few-shot classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526518.

[14] H. Liu, M. Zhang, Z. Di, M. Gong, T. Gao, and A. K. Qin, "A hybrid multitask learning network for hyperspectral image classification with few labels," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5505816.

[15] C. Liu et al., "Contrastive graph convolution network with skip connections for few-shot hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[16] H. Wang, Z. Qiao, W. Zhang, and X. Liu, "Open set cross-domain hyperspectral image classification based on critical reflective learning network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5521114.

[17] W. N. Khotimah, M. Bennamoun, F. Boussaid, L. Xu, and F. Sohel, "Dual-phase framework for few-shot hyperspectral image classification with spatiospectral masked autoencoder and episode training," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4404616.

[18] G. Xu, S. Yang, and Z. Feng, "Dual-semantic graph convolution network for hyperspectral image classification with few labeled samples," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508815.

[19] Z. Li, Z. Xue, Q. Xu, L. Zhang, T. Zhu, and M. Zhang, "SPFormer: Self-pooling transformer for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5502019.

[20] Y. Dong, B. Zhu, X. Yang, and X. Ma, "Deep metric learning based on Brownian covariance representation for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5507813.

[21] N. A. A. Braham, J. Mairal, J. Chanussot, L. Mou, and X. X. Zhu, "Enhancing contrastive learning with positive pair mining for few-shot hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 8509–8526, 2024.

[22] B. Xi et al., "DGSSC: A deep generative spectral–spatial classifier for imbalanced hyperspectral imagery," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1535–1548, Apr. 2023.

[23] W. Wei, L. Tong, B. Guo, J. Zhou, and C. Xiao, "Few-shot hyperspectral image classification using relational generative adversarial network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5539016.

[24] D. Ibañez, R. Fernandez-Beltran, F. Pla, and N. Yokoya, "Masked auto-encoding spectral–spatial transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5542614.

[25] B. Fang, Y. Li, H. Zhang, and J. C.-W. Chan, "Collaborative learning of lightweight convolutional neural network and deep clustering for hyperspectral image semi-supervised classification with limited training samples," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 164–178, Mar. 2020.

[26] Z. Ye, J. Wang, H. Liu, Y. Zhang, and W. Li, "Adaptive domain-adversarial few-shot learning for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5532017.

[27] X. Liu, L. Tong, J. Zhou, and C. Xiao, "Adversarial dilated large-kernel attention networks for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5517218.

[28] Y. Zhao, J. Sun, N. Hu, C. Zai, and Y. Han, "Residual channel attention based sample adaptation few-shot learning for hyperspectral image classification," *Sci. Rep.*, vol. 14, no. 1, p. 26746, Nov. 2024.

[29] F. Liu, W. Gao, J. Liu, X. Tang, and L. Xiao, "Adversarial domain alignment with contrastive learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525720.

[30] M. Karimpour, S. Noori Saray, J. Tahmoresnezhad, and M. Pourmahmood Aghababa, "Multi-source domain adaptation for image classification," *Mach. Vis. Appl.*, vol. 31, no. 6, p. 44, Sep. 2020.

[31] X. Liang, Y. Zhang, and J. Zhang, "Attention multisource fusion-based deep few-shot learning for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8773–8788, 2021.

[32] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[33] M. Ahmad, M. Usama, M. Mazzara, and S. Distefano, "WaveMamba: Spatial–spectral wavelet mamba for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[34] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[35] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," in *Proc. Int. Conf. Mach. Learn.*, 2024, pp. 62429–62442.

[36] G. Wang, X. Zhang, Z. Peng, T. Zhang, and L. Jiao, "S ^{2}mamba: A spatial–spectral state space model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5511413.

[37] S. Zhang, Z. Chen, and F. Zhong, "Cross-domain multimodal feature enhancement hypergraph neural network for few-shot hyperspectral images classification," *Expert Syst. Appl.*, vol. 283, Jul. 2025, Art. no. 127742.

[38] F. Melgani and L. Bruzzone, "Support vector machines for classification of hyperspectral remote-sensing images," in *Dig. Int. Geosci. Remote Sens. Symp.*, Toronto, ONT, Canada, Jun. 2002, pp. 506–508.

[39] X. He, Y. Chen, and Z. Lin, "Spatial-spectral transformer for hyperspectral image classification," *Remote Sens.*, vol. 13, no. 3, p. 498, Jan. 2021.

[40] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[41] H. Sun, X. Zheng, X. Lu, and S. Wu, "Spectral–spatial attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3232–3245, 2019.

[42] B. Qin, S. Feng, C. Zhao, W. Li, R. Tao, and W. Xiang, "Cross-domain few-shot learning based on feature disentanglement for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5514215.

[43] C. Shi, W. Liu, L. Fang, Z. You, Q. Miao, and C.-M. Pun, "Few-shot learning based on multilevel contrast for cross-domain hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5522918.

[44] H. Liu, J. He, X. Zhang, and Z. Li, "Momentum-enhanced dual-prototype learning framework for robust few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5534816.

[45] Z. Zheng, Y. Zhong, A. Ma, and L. Zhang, "FPGA: Fast patch-free global learning framework for fully end-to-end hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 1–15, Aug. 2020.

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